

Anomalies and False Rejections

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We use information from over 2 million trading strategies randomly generated using real data and from strategies that survive the publication process to infer the statistical properties of the set of strategies that could have been studied by researchers. Using this set, we compute t -statistic thresholds that control for multiple hypothesis testing, when searching for anomalies, at 3.8 and 3.4 for time-series and cross-sectional regressions, respectively. We estimate the expected proportion of false rejections that researchers would produce if they failed to account for multiple hypothesis testing to be about 45%. (*JEL* C12, G12)

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Researchers in finance, like those in other academic fields, have long been worried about the possibility that a substantial number of reported discoveries

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might be false.¹ The problem might be particularly severe for empirical asset pricing studies that investigate cross-sectional predictability in stock returns. Part of the difficulty might be the way in which these strategies are discovered or the absence of well-specified research practices (Harvey 2017).

Nevertheless, there exists a fundamental statistical issue: when many hypotheses are tested, some will appear significant at conventional thresholds from single hypothesis testing (SHT) even if they are true under the null. We refer to such rejections as false rejections or false discoveries under SHT. Although independently carried out by many researchers, the predictability studies share a common question (i.e., are (expected) stock returns predictable?) and, almost invariably, at least one common data set (i.e., CRSP). Accounting for multiple hypotheses testing (MHT) on a common question and data set thus might offer not only a way to determine the severity of the false rejection problem but also a solution to it.

Despite a number of recent studies that propose the application of MHT methods, the magnitude of the false discovery problem and, closely related, the thresholds that should be applied to test statistics to reduce false discoveries, remain a matter of debate. For example, drawing from a meta-analysis of 296 published discoveries, Harvey, Liu, and Zhu (2016) suggest a MHT *t*-statistic threshold higher than three, but warn that there are very good reasons why this number might be too low for studies that are not grounded in theory. Using the Benjamini and Yekutieli (2001) procedure to obtain an MHT threshold, they calculate that 80 of the 296 discoveries (i.e., 27%) are rejected under SHT, but not under MHT. We refer to these rejections as “single, but not multiple” (SnM) rejections.

On the other hand, in a set of randomly generated trading strategies and using a bootstrap approach (but not formal MHT methods), Yan and Zheng (2017) settle on a rather low statistical threshold that implies a very small number of false discoveries and thousands of informative signals. The conflicting evidence stems from two sources: the method used to determine an MHT adjusted threshold and the set of trading strategies studied. Harvey, Liu, and Zhu (2016) and Yan and Zheng (2017) fundamentally differ from one another on both counts. For example, Yan and Zheng (2017) use a laboratory of over 18,000 randomly generated signals and find that a vast majority of them can be rejected using bootstrap method of Fama and French (2010), but not a formal MHT procedure. Harvey and Liu (2019) show that the bootstrap methods have severe size biases that lead to large overrejections.

The problem of choosing an appropriate MHT method can be clarified by weighing the underlying assumptions in a simulation context. Because asset pricing studies rely on data sets where the data exhibit serial and cross-sectional

¹ Leamer (1983, p. 43) complains of the “fumes which leak from our computing centers” and argues that it is critical that the fragility of results be studied in a systematic way to establish or alter beliefs. In fact, Harvey, Liu, and Zhu (2016) argue that “. . . most claimed research findings in financial economics are likely false.”

dependence, employing a method robust to these features is preferable. Using simulation evidence, we suggest the adoption of the procedure advanced by Romano and Wolf (2007) and Romano, Shaikh, and Wolf (2008), RSW henceforth.

Even after having identified a well-suited MHT method, resolving the issue of which (correct) set of trading strategies to study is relatively more complicated. As Harvey, Liu, and Zhu (2016) Liu, note, in an ideal setting, one would apply MHT to all the strategies that researchers have attempted to test (say, set $\mathcal{S}^{\mathcal{R}}$), and not just those that eventually become public (say, set $\mathcal{S}^{\mathcal{P}}$) because they ended up in the tails of the distribution of t -statistics (i.e., those for which the null was rejected and, thus, made it to a circulated paper). Application of MHT to the set $\mathcal{S}^{\mathcal{P}}$, which does not contain all the strategies that are obviously false, would produce a low threshold for statistical significance, and a small proportion of SnM (i.e., single, but not multiple) rejections.

One feasible alternative is to consider a large set of randomly generated strategies (say set $\mathcal{S}^{\mathcal{E}}$ that an econometrician draws) that mechanically contains most of the strategies that researchers have attempted but that were not significant. Unfortunately, the set $\mathcal{S}^{\mathcal{E}}$ also contains many strategies that researchers would never study because their economic foundation is not immediately justifiable. Assuming that economic reasoning gives researchers an advantage in filtering out strategies for which the null of no predictability is true, the set $\mathcal{S}^{\mathcal{E}}$ is then biased in the opposite direction: it produces MHT adjusted thresholds that are too high and, consequently, a very high estimate of SnM rejections. We generate such a large set (of approximately 2.4 million signals) and apply the RSW procedure to t -statistics of long-short portfolios abnormal returns from a six-factor model and Fama and MacBeth (1973) (FM) slope coefficients. We find SnM rejections to be 98% in this set $\mathcal{S}^{\mathcal{E}}$.

In summary, the proportion of false rejections is an unobservable quantity that cannot be directly estimated from the data. It can be only indirectly inferred by making explicit assumptions about the data-generating process and about the ability of researchers to filter out strategies that are economically unsound.

We propose a statistical model that allows us to use the information contained in the sets $\mathcal{S}^{\mathcal{E}}$ and $\mathcal{S}^{\mathcal{P}}$ to extract the statistical properties of the set $\mathcal{S}^{\mathcal{R}}$, and hence indirectly infer the proportion of false rejections under SHT. Signals are drawn from a mixture distribution wherein some signals are informative, and others are not. The critical assumption is that the probability of an informed signal in the set $\mathcal{S}^{\mathcal{R}}$ is proportional by a factor larger than one to the equivalent probability in the set $\mathcal{S}^{\mathcal{E}}$. The intuition behind this assumption is that researchers can use economic reasoning to select better signals than those randomly selected by the econometrician. Hence, researchers are endowed in the model with a higher ability to draw informative signals. We calibrate the model, by matching quantities observable in the data and in the simulated economy produced by the model.

With the set $\mathcal{S}^{\mathcal{R}}$ in hand, we calculate RSW-adjusted thresholds that allow a researcher to control the false discovery proportion (FDP) (at the 5% level). In

our application, MHT-adjusted thresholds for t -statistics of time-series alphas and cross-sectional FM regression slopes are 3.8 and 3.4, respectively. How strict is the 3.4 threshold? Say that many researchers conduct independent tests of an economic relation that is not true. These researchers draw t -statistics from the standard normal distribution. A statistical threshold of 3.4 implies that at least 1,028 tests would need to be attempted to have 50% probability that at least one of the t -statistics crosses the 3.4 threshold. To cross a threshold of 3.8 a total of 4,790 tests would have to be attempted. For comparison, 14 attempts would give the same probability of crossing the SHT threshold of 1.96. Thus, the high MHT thresholds offer very strong protection against false discoveries, protection that is much stronger than that afforded by SHT. The protection also considerably increases when moving from 3.4 to 3.8.

There are economic reasons why time-series alphas and FM cross-sectional regressions provide different tests of the same hypothesis (see, e.g., Fama and French 2008). In our implementation, we obtain different thresholds because cross-sectional regressions are less prone to produce false rejections, and thus require a lower threshold to guarantee the same FDP control. However, we recommend researchers to do both tests and apply the two different MHT thresholds. In fact, imposing dual hurdles decreases the probability of false rejections under both SHT and MHT.

While our thresholds are relatively stable with respect to different factor models, different choices of samples, as well as alternative assumptions about returns and signals, we caution that these thresholds are obtained by applying the RSW procedure to a specific context of empirical asset pricing studies that are concerned with abnormally profitable trading strategies. In different applications, our thresholds would not necessarily guarantee the same control of FDP. Hence, we suggest that researchers calculate their own thresholds using the same RSW procedure and, at a minimum, accounting for all specifications that they have tried. Further, our statistical approach (indeed any MHT procedure) addresses only the multiplicity problem that naturally arises in the research process, and hence is of little use in aligning researchers' incentives. We refer readers to Harvey (2017) for a discussion of the publication process and the incentives therein.

Our statistical model also allows us to quantify the proportion of false discoveries to be about 45%. We interpret our estimate as the proportion of false discoveries that one should expect if finance researchers do not adopt MHT adjustments and, instead, rely only on SHT thresholds. It is important to note that our estimate of false rejections is based only on the assumption that published strategies have statistically significant alpha and FM coefficient t -statistics under SHT. In typical published research, authors need a theory/motivation/story to have a better chance of getting published. These additional qualitative hurdles are not taken into consideration in our purely statistical framework.

To contextualize our estimate of the proportion of false discoveries, we turn to the literature. Linnainmaa and Roberts (2018) report that, of the 36 strategies that they study, 20 strategies have an insignificant Fama and French (1993) three-factor alpha in the pre-discovery period. This represents a false rejection rate of 56%. McLean and Pontiff (2016), find a post publication decay of 58% in the average return of 97 trading strategies. They attribute 26% to statistical decay and 32% from publication-informed trading. If returns from all false strategies decayed to zero and those from all true strategies did not decay, then this would imply an estimate of false rejections of 26%. Our estimate, which is based on in-sample analyses, is in between the out-of-sample estimates of McLean and Pontiff and those of Linnainmaa and Roberts. Thus, contrary to what Martin and Nagel (2019) suggest, in-sample and out-of-sample tests provide approximately the same information, as long as the in-sample analysis is corrected for MHT.

Our research echoes the increasing skepticism about the validity of many research findings in a variety of fields. While the findings on the lack of replicability in medical research by Ioannidis (2005) are widely cited, the economics profession has also made an effort to tackle this problem. Dewald, Thursby, and Anderson (1986), McCullough and Vinod (2003), and Chang and Li (2018) also report disappointing results from the replication of economics papers. The use of replication in finance is less widespread, with McLean and Pontiff (2016) being a notable exception.

Our paper also joins the list of the growing finance literature that studies the proliferation of discoveries of abnormally profitable trading strategies and/or pricing factors and its relation to data-snooping biases in finance. See Lo and MacKinlay (1990), Foster, Smith, and Whaley (1997), Conrad, Cooper, and Kaul (2003), and Karolyi and Kho (2004) for early work emphasizing statistical biases in hypothesis testing. The question of whether the profitability of published strategies survives the test of time is studied in Schwert (2003), Chordia, Subrahmanyam, and Tong (2014), McLean and Pontiff (2016), Linnainmaa and Roberts (2018), and Hou, Xue, and Zhang (2018). Toward the turn of the century, more formal statistical approaches were developed and applied to the problem of evaluating multiple strategies. See, for example, Sullivan, Timmermann, and White (1999), White (2000), and Romano and Wolf (2005). The MHT approach has been more recently applied to financial settings in Barras, Scaillet, and Wermers (2010), Harvey, Liu, and Zhu (2016), and Harvey and Liu (2019) and emphasized in the presidential address of Harvey (2017). Chen and Zimmermann (2018) offer a different Bayesian perspective, by estimating the publication bias in the context of a model that describes the impact of the publication process on experiments designed by researchers. Later in the paper, we relate our results to Chen (2019) who studies 156 strategies and concludes that no major adjustment to statistical thresholds is required.

1. Multiple Hypotheses Testing

Consider a multiple testing situation in which a researcher explores S hypotheses using a particular data set. S_0 (S_1) of these hypotheses are true (false) under the null, H_0 . Each hypothesis is evaluated at a confidence level α (e.g., 5%).² A total of R hypotheses are rejected. The breakup of rejections and nonrejections is organized in the following table:

	H_0 not rejected	H_0 rejected	Total
H_0 true	T_0	F_1	S_0
H_0 false	F_0	T_1	S_1
	$S - R$	$R = F_1 + T_1$	S

Overall, F_1 hypotheses are falsely rejected, and F_1/R is the FDP. F_0 hypotheses should have been rejected but are not, and $F_0/(S - R)$ is the false nondiscovery proportion (FNDP). As S grows, so does F_1 , making the probability of making at least one false discovery (i.e., the rate of Type I errors) increase rapidly. For example, at a conventional significance level of 5%, if all strategies are true under the null and mutually independent, the probability of making at least one false discovery is $1 - 0.95^{10} = 40\%$ when ten hypotheses are tested, and over 99% when 100 hypotheses are tested.

MHT methods control how large the probability of making false rejections is. Three broad approaches differ in the statistical definition of what is actually controlled: familywise error rate (FWER), false discovery rate (FDR), and false discovery proportion (FDP). We discuss the basic intuition behind these approaches and relegate the details to the appendix. See also Harvey, Liu, and Saretto (2019) for a comprehensive review of different MHT methods and their applications to finance.

1.1 Familywise error rate

The most basic idea in MHT is to control the FWER, defined as the probability of rejecting at least one of the true null hypotheses:

$$\text{FWER} = \text{Prob}(F_1 \geq 1).$$

A testing method is said to control the FWER at a significance level α if $\text{FWER} \leq \alpha$. Four main approaches can be employed to control FWER: (1) Bonferroni (1936), (2) Holm (1979), (3) Bootstrap reality check of White (2000), and (4) StepM method of Romano and Wolf (2005). The Bonferroni correction does not require any assumptions, the second and third methods assume independence between hypotheses, and the fourth method is valid under

² [CG1][a2]The use of α to denote both the significance level and abnormal returns from a factor model is standard. We hope that this does not cause any confusion and the usage is clear from the context.

arbitrary dependence in data. MHT methods that control FWER are particularly strict because they try to keep the number of false discoveries at the lowest level (i.e., one false discovery).

All these methods owe their origin to the Bonferroni (1936) adjustment. The logic behind Bonferroni is that if each hypothesis is tested at some significance level α^* , then each one hypothesis gets erroneously rejected with probability α^* . Because S hypotheses are tested, the expected number of false rejections is $E(F_1) = S \times \alpha^*$. Because $\text{Prob}(F_1 \geq 1) \leq E(F_1)$, to guarantee that $\text{FWER} \leq \alpha$, one should set $\alpha^* = \alpha/S$. As the number S of hypotheses being tested increases, the correction becomes more and more stringent, leading to very high t -statistic thresholds for rejection. To varying degrees, all procedures that control FWER suffer from the same problem.

One might be willing to tolerate a larger number of false rejections, in order to discover a larger number of true rejections. For example, the k -FWER control allows a researcher to allow F_1 to be as large as k , as opposed to one. Formally,

$$k\text{-FWER} = \text{Prob}(F_1 \geq k).$$

Therefore, a possible solution to relax the stringent constraints imposed by (1)-FWER is to adopt a k -FWER procedure.

1.2 False discovery proportion

An alternative to controlling the “number” of false rejections is to control the “proportion” F_1/R of false rejections (FDP). Formally, a multiple testing procedure controls FDP at proportion γ and significance level α if

$$\text{Prob}(\text{FDP} \geq \gamma) \leq \alpha.$$

FDP is a random variable, so this condition essentially imposes a restriction on the left tail of its distribution. As such, FDP control guarantees that, in any application, the realized FDP cannot (statistically) exceed a particular threshold γ .

Many algorithms have been proposed to control FDP (see Farcomeni 2008 for a general review). We implement the RSW method proposed by Romano and Wolf (2007) and Romano, Shaikh, and Wolf (2008). This procedure runs a sequence of k -FWER procedures with $k=1,2$, etc. Say that, at a generic step k , the total number of rejections is R_k . Step k corresponds to k -FWER, so we can be sure that at most k false rejections have taken place so far. This means that one additional rejection in the next step will make the FDP equal to $k/(R_k+1)$. To control the FDP at the desired significance level γ , we undertake the next step only if $k/(R_k+1) \leq \gamma$, or, equivalently, only if $R_k \geq k/\gamma - 1$. The last inequality determines the stopping condition of the algorithm.

One desirable property of the RSW algorithm is that it is based on a resampling method. As long as the resampling method preserves the dependence structure in the data, it also allows FDP control under the same arbitrary dependence structure. The main disadvantage of the RSW algorithm

is that, because of its recursive nature and because it relies on resampling the data, it is computationally burdensome.

In our implementation, we rely on a bootstrap procedure that maintains the cross-sectional and the time-series structure in the data. Appendix Section A.5 provides the details of the bootstrap procedure. Here, we note that our approach is inspired by the work of Kosowski et al. (2006), Fama and French (2010), and Yan and Zheng (2017).

1.3 False discovery rate

An alternative to controlling the tail behavior of FDP is to control its expected value, the FDR. Formally, a multiple testing method controls FDR at level δ if

$$\text{FDR} \equiv \text{E}(\text{FDP}) \leq \delta,$$

where δ is a tolerance level imposed by the researcher.

The two main FDR control methods that we consider are the Benjamini and Hochberg (1995) method and its extension by Benjamini and Yekutieli (2001). We label these methods BH and BHY, respectively. We note that BH assumes independent tests while BHY accounts for (partial) dependence in the data. As more general assumptions imply a loss of power, BHY is less powerful than BH (i.e., BHY produces higher thresholds than BH).

In BH and BHY, one orders the hypotheses by their p -values so that the first hypothesis is the most significant. Say that j is the generic index of an ordered hypotheses (i.e., j is the j -th most significant hypothesis and has p -value of p_j). If we rejected j hypotheses, the expected number of false rejections would be $S \times p_j$ and, therefore, $(S \times p_j)/j$ would be the realized FDR. The procedure stops at j^* such that the corresponding FDR is less than δ .

One desirable property of the BH and BHY algorithms is that they are computationally easy to implement. An important conceptual limitation is that, in a given application (i.e., in a given data set), the realized FDP can be far away from the average (FDR) and, hence, from the tolerance level δ (see Genovese and Wasserman 2006 for a more detailed discussion). In other words, specifying the FDR control δ at 5%, say, could still lead to a *realized FDP in a particular data set* being far above 5%, with the incidence of such cases increasing the further away we move from the independence assumption in the tests. Therefore, FDR control is better suited for cases in which a researcher can analyze a large number of data sets, allowing one to make confidence statements about the average realized FDP across such data sets.³

2. Monte Carlo Simulations

In this section, we first clarify the realized statistical properties of MHT methods. We give preference to some over others and, consequently, suggest

³ We thank Michael Wolf for clarifying this important difference for us.

relying only on those methods for inference. Second, we show in a controlled experiment that the proportion of false discoveries is greatly reduced by implementing a dual hurdle on alphas and FM coefficients.

2.1 Simulation details

2.1.1 Data-generating process. Stocks: There are N stocks whose returns are generated from a linear factor model:

$$R_{it} = \alpha_i + \beta_i' F_t + \epsilon_{it}. \quad (1)$$

Alphas are drawn from a normal distribution $\mathcal{N}(0, \sigma_\alpha^2)$. The random shocks ϵ are distributed as $\mathcal{N}(0, \sigma_\epsilon^2)$ where $\sigma_\epsilon = 15.1\%$ to match the average monthly standard deviation of residuals from stock return time-series regressions. We set $N = 2,000$ to roughly equal the monthly average number of stocks for which we can generate a signal in the real data (2,383), and $T = 500$ to roughly equal the number of months in the data (516) (data construction details are in Section 3). The simulated empirical distribution of factors and factor sensitivities are matched to the corresponding empirical quantities in the data. In particular, we simulate a six-factor model that mimics the statistical properties of the five Fama and French (2015) factors augmented with the momentum factor. Each month, we draw the factor returns, F_t , from a multivariate normal distribution with means and covariance matrix that match those of the actual distribution of factor returns. For each stock i , we draw the 6×1 vector of betas, β_i , from a multivariate normal distribution with means and covariance matrix matched to that of the cross-sectional distribution of betas from the actual data. Signals: In each period t , the econometrician draws S noisy signals from a set $\mathcal{S}^\mathcal{E}$. Some of the signals contain information about returns and some do not. We can think about a signal as a variable constructed from accounting data. Some signals (e.g., the book-to-market ratio) are informative in the sense that they carry some cross-sectional predictability about stocks' abnormal returns (i.e., α_i), whereas others (e.g., the ratio of debt due in one year to depreciation) are not. We draw an informative signal with probability π . After drawing a signal, its realizations corresponding to stock i in time period t are given by

$$s_{it} = \begin{cases} \alpha_i + \eta_{it}^s & \text{if the signal is informative,} \\ \eta_{it}^s & \text{if the signal is uninformative.} \end{cases} \quad (2)$$

Even informative signals, however, are not perfect, and $\eta^s \sim \mathcal{N}(0, \sigma_\eta^2)$ is signal-specific noise. The signals' ability to predict returns depends on the signal-to-noise ratio, $\pi \times \sigma_\alpha^2 / \sigma_\eta^2$. Since signals in the real world are correlated (many signals come from the same data sources, such as balance sheet and cash flow statements), so we allow for a constant correlation ρ among signals. Unless otherwise stated, we set $S = 10,000$ and $\pi = 5\%$.

2.1.2 Portfolios and cross-sectional regressions. We analyze the simulated data in a similar way as we do with the real data. Every month, we sort stocks

into deciles based on signal s and hold these portfolios for 1 month. The hedge portfolio goes long in decile 10 and short in decile 1. We run a time-series regression of the hedge portfolio returns, R_{st} , based on signal s , on factor returns F_t and record the t -statistic of the alpha, t_{α_s}

$$R_{st} = \alpha_s + \beta'_s F_t + u_{st}. \quad (3)$$

We obtain S portfolios and corresponding number of t_{α_s} . Every period t we also run a univariate cross-sectional regression of risk-adjusted returns, $(R_{it} - \hat{\beta}'_i F_t)$ as in Brennan, Chordia, and Subrahmanyam (1998), on signal s_{it}

$$R_{it} - \hat{\beta}'_i F_t = \lambda_{0t} + \lambda_{st} s_{it} + \psi_{it},$$

and compute the usual Fama and MacBeth (1973) t -statistic, t_{λ_s} .

2.2 Properties of different MHT methods

Table 1 reports thresholds, rejection rates, the average and the standard deviation of FDP, and the average false nondiscovery proportion (FNDP), which is inversely related to power. We tabulate results obtained from applying MHT methods to the alpha t -statistics obtained from our simulations. We adopt $\alpha=5\%$ statistical significance level, $\gamma=5\%$ for FDP control, and $\delta=5\%$ for FDR control. Each entry in the table corresponds to the average or standard deviation over $K=1,000$ repetitions. We can think of each simulated economy as an independent data set that is used to study the informativeness of the S signals.

The baseline parameters for the simulation are $N=2,000$, $T=500$, $\pi=5\%$, $\sigma_\eta=20\%$, $\sigma_\alpha=2\%$, $\rho=0$, and $S=10,000$. Different panels of Table 1 vary one parameter while keeping the others fixed. Panel A varies the signal-to-noise ratio by varying σ_α from 1.5% to 2.5%, panel B varies the correlation of signals ρ from zero to 20%,⁴ panel C varies the proportion of informative signals π from zero to 30%, and panel D varies the number of signals S from 5,000 to 500,000. To avoid clutter, we do not discuss each panel separately but rather discuss only the main conclusions of the analysis.

First, the FWER control is very strict and imposes very high statistical thresholds, that deliver very few rejections and poor power when the signal-to-noise ratio is low (see panel A). Methods that control FWER are also not adaptive to changes in many of the parameters. For example, increasing the fraction π of informative signals does not change the thresholds (see panel C). Rather, these methods depend primarily on the number of signals S ; higher S (see panel D) mechanically increases the thresholds, reduces rejections, and decreases power. Changing correlation (panel B) does not affect the performance of these methods in an appreciable way. We conclude that FWER control is less suitable for finance applications that exhibit poor signal-to-noise ratio.

⁴ Green, Hand, and Zhang (2017) show that the absolute correlation among strategies that are promoted as being very profitable is only 9%.

Second, of procedures that control FDR and FDP, we notice the following: (a) BHY has the lowest rejections while BH has the highest rejections, with RSW somewhere in between (see panel A). (b) the realized FDR (i.e., $E[FDP]$) is close to 5% for BH but only 0.5% for BHY; thus BHY is too strict. The realized FDR for RSW is well below 5%. This is to be expected because FDP control at $\gamma=5\%$ implies a realized FDR of below 5%. (c) As mentioned in Section 1.3, one drawback of FDR control is it does not guarantee that the realized FDP will be below 5% in a particular data set. This is reflected in the relatively high standard deviation (across simulations) of FDP for BH; BHY still manages to have low $SD[FDP]$ because of its high thresholds and low rejections (again see panel A). High correlation of signals creates the biggest problems for BH (see panel B). (d) Power ($=1-E[FNDP]$) is directly associated with the strength of the signal. RSW sits in the middle between BH and BHY in terms of power. Third, thresholds do not mechanically increase with an increase in signals for all MHT methods (see panel D). Thresholds do increase for methods that control FWER; this fact may have contributed to the erroneously held popular belief that MHT thresholds will keep increasing as the profession keeps discovering more strategies. However, thresholds are relatively insensitive for methods that control FDR and FDP. If anything, a higher number of tests leads to a slight gain in efficiency, as evidenced by the lower $SD[FDP]$, for

Table 1
MHT properties in simulations

A. Signal-to-noise ratio						B. Correlation between signals					
σ_α	Bonf	Holm	BH	BHY	RSW	ρ	Bonf	Holm	BH	BHY	RSW
Thresholds						Thresholds					
1.5	4.6	4.6	3.1	3.8	3.5	0	4.6	4.5	3.0	3.8	3.4
2.0	4.6	4.4	3.0	3.8	3.4	10	4.6	4.5	3.0	3.9	3.5
2.5	4.6	4.4	3.0	4.1	3.4	20	4.6	4.5	3.0	4.0	3.6
Rejections						Rejections					
1.5	1.8	1.8	4.6	3.4	3.7	0	5.0	5.0	5.2	5.0	5.1
2.0	5.0	5.0	5.2	5.0	5.1	10	5.0	5.0	5.3	5.0	5.1
2.5	5.0	5.0	5.2	5.0	5.1	20	5.0	5.0	5.3	5.0	5.0
E[FDP]						E[FDP]					
1.5	<0.1	<0.1	4.8	0.5	1.0	0	<0.1	<0.1	4.7	0.5	1.2
2.0	<0.1	<0.1	4.7	0.5	1.2	10	<0.1	<0.1	4.8	0.5	1.0
2.5	<0.1	0.2	4.7	0.5	1.2	20	<0.1	<0.1	4.8	0.5	0.8
SD[FDP]						SD[FDP]					
1.5	0.1	0.1	1.0	0.4	0.5	0	<0.1	0.1	1.0	0.3	0.5
2.0	<0.1	0.1	1.0	0.3	0.5	10	<0.1	<0.1	2.3	0.5	0.8
2.5	<0.1	0.1	1.0	0.3	0.5	20	0.1	0.1	4.3	0.8	1.2
E[FNDP]						E[FNDP]					
1.5	63.9	63.7	12.7	33.0	25.8	0	0.3	0.3	<0.1	<0.1	<0.1
2.0	0.3	0.3	<0.1	<0.1	<0.1	10	0.2	0.2	<0.1	<0.1	<0.1
2.5	0	0	0	0	0	20	0.3	0.3	<0.1	<0.1	<0.1

(Continued)

Table 1
(Continued)

C. Proportion of informative signals						D. Number of signals					
π	Bonf	Holm	BH	BHY	RSW	S	Bonf	Holm	BH	BHY	RSW
Thresholds						Thresholds					
0	4.6	4.5	3.0	3.8	3.4	5,000	4.4	4.4	3.0	4.1	3.5
10	4.6	4.5	2.8	3.5	3.2	10,000	4.6	4.5	3.0	3.8	3.4
20	4.6	4.5	2.6	3.3	2.8	100,000	5.0	5.0	3.0	3.7	3.4
30	4.6	4.5	2.4	3.2	2.6	500,000	5.2	5.2	3.0	3.7	3.4
Rejections						Rejections					
0	5.0	5.0	5.2	5.0	5.1	5,000	5.0	5.0	5.2	5.0	5.0
10	10.0	10.0	10.5	10.0	10.2	10,000	5.0	5.0	5.2	5.0	5.1
20	19.9	20.0	20.8	20.1	20.4	100,000	4.9	4.9	5.2	5.0	5.1
30	29.9	29.9	31.1	30.1	30.6	500,000	4.9	4.9	5.2	5.0	5.1
E[FDP]						E[FDP]					
0	<0.1	<0.1	4.7	0.5	1.2	5,000	<0.1	<0.1	4.7	0.5	0.9
10	<0.1	<0.1	4.4	0.5	1.5	10,000	<0.1	<0.1	4.7	0.5	1.2
20	<0.1	<0.1	3.9	0.4	1.8	100,000	<0.1	<0.1	4.7	0.4	1.2
30	<0.1	<0.1	3.5	0.4	2.1	500,000	<0.1	<0.1	4.7	0.4	1.2
SD[FDP]						SD[FDP]					
0	<0.1	0.1	1.0	0.3	0.5	5,000	0.1	0.1	1.3	0.5	0.7
10	<0.1	<0.1	0.7	0.2	0.5	10,000	<0.1	0.1	1.0	0.3	0.5
20	<0.1	<0.1	0.4	0.1	0.4	100,000	<0.1	<0.1	0.4	0.1	0.3
30	<0.1	<0.1	0.3	0.1	0.4	500,000	<0.1	<0.1	0.3	0.1	0.3
E[FNDP]						E[FNDP]					
0	0.3	0.3	<0.1	<0.1	<0.1	5,000	0.2	0.2	0	<0.1	<0.1
10	0.3	0.3	<0.1	<0.1	<0.1	10,000	0.3	0.3	0	<0.1	<0.1
20	0.3	0.2	<0.1	<0.1	<0.1	100,000	1.2	1.1	<0.1	<0.1	<0.1
30	0.3	0.2	0	<0.1	0	500,000	1.8	1.9	<0.1	<0.1	<0.1

Data are generated for N stocks and S signals as described in the text. Stocks alphas are normally distributed with mean zero and variance σ_α^2 . A fraction, π , of signals are informative, and σ_η is the noise in the signal. Signals are correlated with constant correlation of ρ . The Bonf(erroni) and Holm methods control FWER; the BH and BHY methods control FDR; and RSW controls FDP. We report thresholds, rejection rates, the average and the standard deviation of FDP, and the average false nondiscovery proportion (FNDP) for alpha t -statistics, where FDP and FNDP are calculated for each simulation based on the knowledge of informative signals. We use $\alpha=5\%$ statistical significance level, $\gamma=5\%$ for FDP control, and $\delta=5\%$ for FDR control. The statistics below are reported for $K=1,000$ repetitions. Unless otherwise noted, $N=2,000$, $T=500$, $\pi=5\%$, $\sigma_\eta=20\%$, $\sigma_\alpha=2\%$, $\rho=0$, and $S=10,000$. Panel A varies the signal-to-noise ratio by changing σ_α ; panel B varies the correlation ρ among signals; panel C varies the proportion π of informative signals; and panel D varies the number S of signals.

these methods. We conclude that (a) having a large number of strategies does not impose any bias on methods that control FDR and FDP and (b) there is a slight efficiency advantage to having more strategies over having fewer strategies.

In summary, the evidence suggests a preference for the RSW method for three reasons. First, FWER control is not appropriate for finance applications as it lacks power and is mechanically affected by the number of hypotheses being tested. Second, RSW is in between the two FDR control methods in terms of both size and power. Third, as RSW is meant to control the tail behavior of

FDP, it avoids the concerns that controlling FDR exposes a researcher to the possibility that the realized FDP varies significantly across applications. Thus, our suggestion is to adopt RSW.

2.3 Dual hurdles

Portfolio sorts and FM regressions both have advantages and disadvantages (Fama and French 2008). The alphas of the long-short portfolio effectively consider the efficacy of the strategy in only 20% of the sample while FM regressions consider a signal's ability to predict returns in the entire cross-section of stocks. On the other hand, FM regressions impose linearity between signals and returns while portfolio sorts are a nonparametric way of uncovering the relationship. Therefore, many papers in the finance literature report both alpha and FM t -statistics.

Statistically, imposing dual hurdles can be seen as a way to improve the size of the tests. The probability of a strategy having statistically significant alpha and FM coefficient by luck is lower than having either of them (if alphas and FM coefficients are not perfectly correlated).

Accordingly, we next compare rejection rates and realized FDR (i.e., $E[\text{FDP}]$) obtained from applying SHT and RSW to alpha, FM, and *both* alpha and FM t -statistics. We set $N = 2,000$, $T = 500$, $S = 10,000$, $\rho = 0$, and $\sigma_\eta = 20\%$. We set π at either 0% or 5% and vary σ_α from 1.5% to 2.5%.

Table 2 shows that the rejection rates for SHT for either alpha or FM are close to 5% for $\pi = 0$ and close to 10% for $\pi = 5\%$. The rejection rates increase with signal-to-noise ratio. These numbers are to be expected, because SHT not only rejects 5% of the strategies that should be rejected (assuming very good power) but also rejects 5% of the strategies from the null (false rejections). Imposing a dual hurdle reduces the proportion of rejections to almost half. SHT still rejects 1.5% strategies with $\pi = 0$ and about 6% strategies with $\pi = 5\%$. Rejection rates based on RSW for either alpha or FM are low to begin with (lower than 5%) but a dual hurdle reduces the rejection rates further.

A sharper picture is obtained by looking at the realized FDR. SHT rejects roughly twice as many strategies as π , so the realized FDR under SHT is close to 50% for either alpha or FM. Imposing the dual hurdle reduces the FDR by half. Thus, imposing a dual hurdle helps in curbing the false rejection rate to about 25% for SHT, which might still be deemed too high. MHT helps in improving size. The dual hurdle reduces the realized FDR to about 0.2% when asking strategies to have t -statistics that cross both alpha and FM thresholds. In other words, MHT is designed to limit false rejections but permits some false rejections (to improve power). Imposing a dual hurdle reduces the chances of false rejections further. In fact, dual hurdles reduce false rejections to even below the prespecified control (of 5%). We are not aware of any statistical method that guarantees FDR or FDP control of a dual hurdle procedure at a

Table 2
Dual hurdles

π	σ_α	Threshold	Rejections		E[FDP]	
		RSW	RSW	SHT	RSW	SHT
Portfolio alpha: t_α						
0	2.0	4.0	<0.1	4.9	—	—
5	1.5	3.5	3.7	9.6	1.1	48.7
5	2.0	3.4	5.1	9.7	1.2	48.2
5	2.5	3.4	5.1	9.7	1.2	48.4
Fama-MacBeth: t_λ						
0	2.0	4.0	<0.1	5.0	—	—
5	1.5	3.2	5.1	9.8	2.7	48.9
5	2.0	3.2	5.1	9.8	2.7	48.9
5	2.5	3.2	5.1	9.8	2.8	48.9
t_α and t_λ						
0	2.0	—	<0.1	1.5	—	—
5	1.5	—	3.7	6.3	0.2	22.0
5	2.0	—	5.0	6.4	0.2	21.8
5	2.5	—	5.0	6.4	0.2	21.9

Data are generated for N stocks and S signals as described in the text. Stocks alphas are normally distributed with mean zero and variance σ_α^2 . A fraction of π signals are informative, and σ_η is the noise in the signal. Signals are correlated with constant correlation of ρ . We set $N=2,000$, $T=500$, $S=10,000$, $\rho=0$, and $\sigma_\eta=20\%$. We set π at either 0% or 5% and vary σ_α from 1.5% to 2.5%. We report results for SHT (with threshold 1.96) and RSW. We use $\gamma=5\%$ for FDP control. All rejection rates are for a significance level of $\alpha=5\%$. We report thresholds, rejection rates, and $FDR \equiv E[FDP]$. We calculate these statistics separately for alpha and the FM coefficient and jointly for alpha and the FM coefficient.

particular level. Nevertheless, given the importance of the issue, we offer some heuristics in Section 4.2.1.

In summary, the results from this section demonstrate that the often-followed practice of dual hurdles helps in reducing the probability of false rejections for all testing methods. Coupled with MHT methods, the practice makes for a very effective tool to guard against false rejections.

3. Data and Trading Strategies

Estimating MHT thresholds and the proportion of false discoveries requires the unobservable set of all strategies studied by researchers, $\mathcal{S}^\mathcal{R}$. We infer the statistical properties of $\mathcal{S}^\mathcal{R}$ from a statistical model, fully described in Section 4, in which a researcher draws signals from a mixture distribution. Some of the signals are informative, and some are not, as in Section 2. The main feature of the model is that the researcher’s skill at drawing informative signals is estimated by asking the model to match several observed quantities. Importantly, if the researcher has no skill, she draws from the same set of randomly generated strategies that an econometrician observes, $\mathcal{S}^\mathcal{E}$. We generate such a set $\mathcal{S}^\mathcal{E}$ from the data and describe its construction below. One can view the set $\mathcal{S}^\mathcal{E}$ as a limit case for the set $\mathcal{S}^\mathcal{R}$.

3.1 Data

Monthly returns and prices are obtained from CRSP. Annual accounting data come from the merged CRSP/Compustat files. We collect all items included in the balance sheet, the income statement, the cash flow statement, and other miscellaneous items for the years 1972 to 2015. We choose 1972 as the beginning of our sample as it corresponds to the first year of trading on Nasdaq that dramatically increased the number of stocks in the CRSP data set. Following convention, we set a 6-month lag between the end of the fiscal year and the availability of accounting information.

We impose several filters on the data to obtain our basic sample. First, we include only common stocks with a CRSP share code of 10 or 11. Second, we require that data for each variable be available for at least 300 firms each month for at least 30 years during the sample period. Third, in FM regressions, we require that data be available for all independent variables (including the variable of interest) for at least 300 firms each month for at least 30 years during the sample period.

Variables (185) that survive the above filters can be used to develop trading signals. Appendix Table A1 lists these variables. We consider several transformations of these variables: growth rates from one year to the next; ratios of all possible combinations of two levels; and ratios of all possible combinations of three variables that can be expressed as the difference of two variables divided by a third variable (i.e., $(x_1 - x_2)/x_3$). We obtain a total of 2,393,641 possible signals. We evaluate trading signals by (a) estimating abnormal performance of the hedge portfolios using a factor model and (b) evaluating the ability of the signal to explain the cross-section of returns.

3.2 Hedge portfolios

We sort firms into value-weighted deciles on June 30 of each year and rebalance these portfolios annually. We discuss results using alternative 2×3 portfolios in robustness Section 5.3. The first portfolio formation date is June 1973 and the last formation date is June 2015. We require a minimum of 30 stocks in each decile (300 stocks in total) in a month to consider that month as having a valid return. The signal is considered to have generated a valid portfolio if there are at least 360 months of valid returns. We consider long-short portfolios only. Thus, we compute a hedge portfolio return that is long in decile ten and short in decile one. We do not know *ex ante* which of the two extreme portfolios has the largest average return, so our hedge portfolios can have either positive or negative average returns. Obviously, obtaining a positive average return for a hedge portfolio that has a negative average return is always possible by taking the opposite position. For expositional convenience, we decide not to force average returns to be positive.

For each trading strategy, we run a time-series regression of the corresponding hedge portfolio returns on the Fama and French (2015) five-factor model augmented with the momentum factor (Carhart 1997) and obtain

the alpha as well as its heteroscedasticity-adjusted t -statistic, t_α . Section 5.3 presents results for the alternative factor models.

3.3 Fama-MacBeth regressions

We estimate the following FM regression each month:

$$R_{it} - \hat{\beta}_i F_t = \lambda_{0t} + \lambda_{1t} X_{it-1} + \lambda_{2t} Z_{it-1} + e_{it}, \quad (4)$$

where X is the variable that represents the signal and Z 's are control variables. We calculate the FM coefficient λ_1 as well its heteroscedasticity-adjusted t -statistic (t_λ). We use the most commonly used control variables: size (i.e., the natural logarithm of the firm's market capitalization), natural logarithm of the book-to-market ratio, past 1- and 11-month return (skipping the most recent month), asset growth, and profitability ratio. Book-to-market is calculated following Fama and French (1993), and asset growth and profitability are calculated following Fama and French (2015). We risk-adjust the returns on the left-hand side of equation (4) following Brennan, Chordia, and Subrahmanyam (1998). We use the same factor model used to calculate hedge portfolio alphas, and calculate full-sample betas, $\hat{\beta}$ s, for each stock. We require at least 60 months of valid returns to estimate the time-series regression beta. Right-hand-side variables are winsorized at the 0.5 and 99.5 percentiles. We discuss results using the ranks of right-hand-side variables in Section 5.3.

In estimating the cross-sectional regressions, we require a minimum of 300 stocks in a month. We also require a minimum of 360 valid monthly cross-sectional estimates during the sample period. Given that we require valid betas for each stock and data on additional control variables, the data requirements for the FM regressions are slightly more stringent than those for portfolio formation.

3.4 Raw returns, alphas, and FM coefficients

Table 3 describes the empirical distributions of t -statistics for monthly raw returns, alphas, and FM coefficients. We calculate the cross-sectional mean, median, standard deviation, minimum, and maximum across signals and report them in panel A. Also reported is the percentage of t -statistics that, in absolute value, cross thresholds of 1.96 and 2.57 corresponding to SHT significance levels of 95% and 99%, respectively.

A large number of strategies have average return t -statistics that exceed conventional statistical significance levels: 10.6% (3.7%) of the strategies have average return t -statistics larger than 1.96 (2.57) (in absolute value). The distribution of alpha and FM coefficient t -statistics reveals even more exceptional performance of strategies than that in raw returns. For example, 23.2% (11.9%) of alpha t -statistics are higher than 1.96 (2.57). Figure 1 depicts the histograms for the cross-section of alpha and FM coefficient t -statistics. The distributions are generally centered around zero and appear to be symmetric. The support for the distributions is consistent with the cross-sectional standard deviations in Table 3.

Table 3
Significance of trading strategies

A. Statistics from SHT							
	Mean	Median	SD	Min	Max	$% t > 1.96$	$% t > 2.57$
Return	−0.08	−0.10	1.21	−7.04	6.72	10.6	3.7
Alpha	−0.19	−0.19	1.62	−7.80	9.01	23.2	11.9
FM	0.07	0.03	1.64	−8.63	8.12	20.0	11.2
B. Statistics from MHT							
	Alpha		FM		Alpha and FM		
Threshold	4.0		3.3		—		
Rejections	1.4		5.5		0.1		
SnM rejections	94.1		72.7		97.9		

Panel A reports the cross-sectional mean, median, standard deviation, minimum, and maximum of the t -statistics of the monthly average return, alpha, and FM coefficients of the 2,393,641 trading strategies, the construction of which is described in the text. The factor model is the Fama and French (2015) five-factor model augmented with the momentum factor. We run FM regressions as in Brennan, Chordia, and Subrahmanyam (1998), and we include size, book-to-market, profitability, asset growth, and 1- and 12-month lagged returns as additional controls beside the variable used to construct the trading strategy. All right-hand-side variables are winsorized at the 0.5 and 99.5 percentiles in these regressions. We also report the percentage of strategies with t -statistics that are larger than 1.96 and 2.57 in absolute value. Panel B shows alpha and FM coefficient statistical thresholds adjusted for MHT as well as the percentage of rejected strategies using the RSW method for FDP control. We use $\alpha = 5\%$ and $\gamma = 5\%$ for FDP control. The sample period is 1972 to 2015.

Next, we implement the RSW procedure and calculate the corresponding rejection rates. We use $\alpha = 5\%$ for significance level and $\gamma = 5\%$ for FDP control. The t -statistic thresholds and the fraction of rejections corresponding to this threshold are reported in panel B of Table 3. We find that the RSW threshold for alpha t -statistic is 4.0 and 1.4% of alphas cross this threshold. Comparing rejection rates from SHT and MHT gives us an estimate of the proportion of SnM rejections. For instance, 23.2% of strategies have $|t_\alpha| > 1.96$ and 1.4% of the strategies have $|t_\alpha| > 3.96$, the threshold imposed by the RSW method. This yields a SnM rejection rate of 94.1% ($= 1 - 1.4/23.2$). FM coefficient thresholds are lower than the corresponding alpha thresholds, so we also find that the FM coefficient SnM rejections are lower than the alpha SnM rejections.

Imposing the dual filter drastically decreases the number of rejections even for SHT. For example, panel A of Table 3 shows that 23.2% (20.0%) of the strategies have alpha (FM coefficient) t -statistic greater than 1.96. The intersection of these two sets gives us only 5.4% of the total number of strategies (number not reported in the table). Thus, consistency across testing procedures play an important role in restricting the set of statistically significant strategies even with SHT. Imposing the constraint of consistency between alpha and FM coefficients, the proportion of SnM rejections is 97.9% using the RSW method for FDP control.

4. Researcher versus Econometrician

We describe the model that allows us to infer the researcher set, $\mathcal{S}^{\mathcal{R}}$, from the econometrician set, $\mathcal{S}^{\mathcal{E}}$, described in the preceding section. We start by

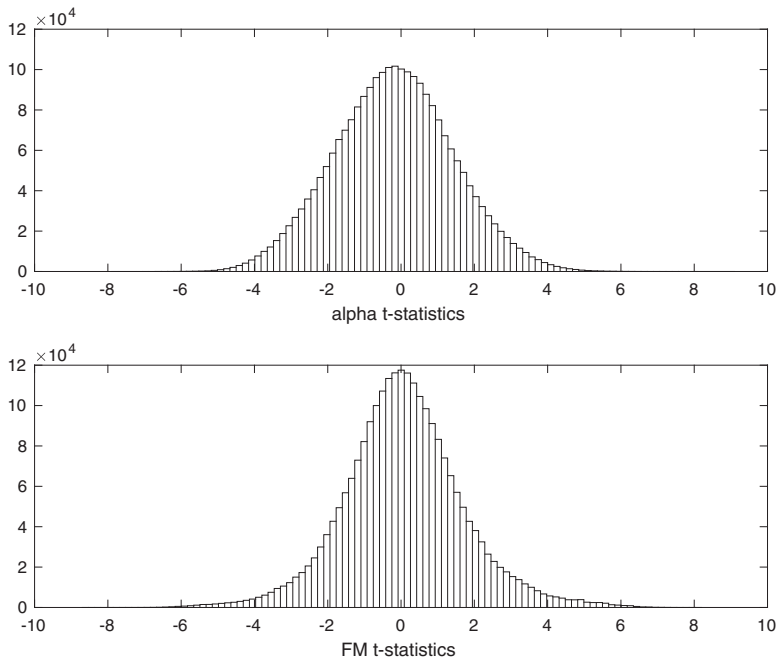


Figure 1
Empirical distributions of portfolios t -statistics

We construct trading strategies as described in the text. The figure shows cross-sectional histograms for alpha t -statistics and Fama-MacBeth coefficient t -statistics. Alphas are computed relative to the Fama and French (2015) five-factor model augmented with a momentum factor. See Table 3 for further details. The sample period is 1972 to 2015.

specifying the different sampling scheme used by the econometrician and the researchers. The econometrician uses a signal s drawn from the set $\mathcal{S}^{\mathcal{E}}$, where a fraction π of signals has the ability to predict returns. We assume that researchers adopt a sophisticated sampling scheme that gives them an advantage over the econometrician. In particular, they draw informative signals with probability $\Omega \times \pi$, where $\Omega \geq 1$ quantifies the advantage that the researchers have over the econometrician.

The idea is that researchers' superior ability comes from the fact that they only sample signals for which they can provide a rationale, by means of a theoretical model or a strong narrative. Consequently, the set $\mathcal{S}^{\mathcal{R}}$ will contain more informative signals and fewer uninformative signals than $\mathcal{S}^{\mathcal{E}}$. Nonetheless, we allow for the possibility that, given the nature of research, sometimes a false narrative also can be provided for signals that come from the null of no predictability; researchers sample these signals with probability $(1 - \Omega \times \pi)$. If $\Omega = 1$, $\mathcal{S}^{\mathcal{R}}$ is the same as $\mathcal{S}^{\mathcal{E}}$. Researchers apply a simple rule and circulate a report only for signals for which they can compute a t -statistic that is in absolute value larger than a threshold, say 1.96. The set of strategies that are

published, $\mathcal{S}^{\mathcal{P}} = \{s \in \mathcal{S}^{\mathcal{R}}, t_s > 1.96\}$, is the simulation counterpart of the set of signals analyzed by Harvey, Liu, and Zhu (2016), who apply the BHY procedure to a set of 296 published factors to produce a SnM rejection rate of 27% (i.e., 80 of 296).⁵

“Correct” RSW thresholds and the proportion of false rejections can be computed using the set $\mathcal{S}^{\mathcal{R}}$ and the knowledge of which signals are informative and which are not. We use the statistical framework characterized by Equations (2) through (5); fix some parameters that can be directly observed in the data; and calibrate those that are unobservable by matching quantities in the real data with counterparts that we can compute in the simulated data produced by the model. With all the parameters in hand, we then calculate the MHT adjusted threshold and proportion of false rejections in the set $\mathcal{S}^{\mathcal{R}}$. The proportion of false rejections can be calculated only in a simulation where we know which strategies are true under the null and which are not.

4.1 Calibration

Calibrated parameters: We only calibrate parameters that are unobservable and set the rest to quantities that are either directly measurable in the data or for which there are close estimates in the literature. We calibrate parameters related to the data-generating process for stocks and signals in Equations (2) and (3) and the researchers’ advantage over econometricians in picking informative signals, σ_{η} , π , and Ω , respectively. Fixed parameters: The distributions of factors, betas, and residuals, that define Equation (2) are matched to their respective counterparts in the data, similar to Section 2. We also fix the standard deviation of the cross-sectional distribution of alphas, σ_{α} , to average cross-sectional standard deviation of the realized stock alphas, 1.9%. To compute this quantity in the real data we proceed as follows: for each stock, we run 90-month rolling window time-series regressions of the stock excess returns on the six-factor model, where we require at least sixty observations. We then compute the cross-sectional standard deviation of the stocks’ alphas for each time period, and finally average across time. As there is only limited information about the correlation structure between strategies, we fix ρ to zero (we return to this issue in the robustness checks presented in the Section 5.1). Target quantities: We choose target quantities that are the most responsible for pinning down the parameters to be calibrated. While all parameters affect all target quantities, the connection between parameters and target quantities simply refers to which target quantity is affected the most by changing one particular parameter.

⁵ Harvey, Liu, and Zhu (2016) calculate higher SnM rejection rates using methods, such as those of Bonferroni (1936) and Holm (1979), that control FWER. However, as we have shown in Section 2.2, these methods are very conservative and suffer from poor power. Harvey, Liu, and Zhu (2016, their appendix A) also estimate missed significant factors with t -statistics between 1.96 and 2.57. Using this enhanced sample, they report higher MHT thresholds, presumably implying higher SnM rejections. See Section 4.2 for a discussion of how our exercise relates to theirs.

The fraction of informative signals π essentially defines the set $\mathcal{S}^{\mathcal{E}}$. We calibrate π by asking the simulation to produce the same fraction of SnM discoveries implied by the RSW thresholds that we observe in the data (i.e., SnM rejections in $\mathcal{S}^{\mathcal{E}} = 97.9\%$).

The factor Ω that defines the superior ability of researchers (relative to the econometrician) to draw informative signals is calibrated by asking the simulation to produce the same proportion of SnM rejections in the set $\mathcal{S}^{\mathcal{P}}$ (i.e., SnM rejections in $\mathcal{S}^{\mathcal{P}} = 27.0\%$).

We calibrate σ_{η} by asking the simulation to match two quantities: the average of cross-sectional standard deviations of residuals from the cross-sectional FM regressions in Equation (5) (i.e., $SD(\psi) = 17.2\%$), and the ratio of (significant) portfolio alphas to (significant) stock alphas. Intuitively, the residuals from the FM regressions are inversely related to the ability of signals to explain the cross-section of returns. For each signal, we first run the FM regression as specified in Equation (5) and obtain $SD(\psi)$ by averaging it across signals.⁶

The noise in signals also directly affects the ability to convert stock alphas into profitable trading strategies. The more noise there is in the (informative) signals, the more difficult it is to construct trading strategies with a large alpha. We construct a measure of “conversion” of stock alphas to portfolio alphas as the ratio of significant portfolio alphas to significant stock alphas, $E(|\text{sign. } \alpha_s|)/E(|\text{sign. } \alpha_i|)$. For stocks, we consider only alphas that are significant at conventional levels. Thus, $E(|\text{sign. } \alpha_i|)$ is simply $E(|\alpha_i| | |t_{\alpha_i}| > 1.96)$. For portfolios, consider only those strategies for which both alpha and FM coefficient are significant. Thus, $E(|\text{sign. } \alpha_s|)$ is $E(|\alpha_s| | |t_{\alpha}| > 1.96 \text{ and } |t_{\lambda}| > 1.96)$. Note here that the requirement of statistical significance is used only as a device to isolate the tails of the distributions (because the average alphas for both stocks and portfolios are close to zero in the simulation and in real data). This quantity is computed as $0.12 (= 0.47\%/3.8\%)$ in the real data. Procedure: In the simulated data, we follow a similar procedure for target quantities, but we also average across the 1,000 simulations. We use a global minimization algorithm (particle swarm of Kennedy and Eberhart 1995) to find the vector of parameters that minimizes the sum of the squared percentage distance of the target quantities. Outputs: After obtaining the calibrated parameters, we run 1,000 simulations of the set $\mathcal{S}^{\mathcal{R}}$. In each simulation we compute the ratio of uninformative signals for which both the alpha and FM coefficient t -statistics are larger than 1.96 (i.e., false rejections) relative to the total number of signals that are rejected by SHT. We report the proportion of false rejections as the average of this number across the 1,000 simulations. We can compute false rejections only because we know exactly which signals are informative and which are uninformative. Similarly, for each simulation we compute the RSW

⁶ There is a slight mismatch in the computation of $SD(\psi)$ between the data and the simulation. This quantity for the data is obtained from Equation (1) but obtained from Equation (5) for the simulations as we do not have the control variables, such as size, book-to-market ratio, and past returns, for the simulations.

Table 4
False rejection rates and adjusted MHT thresholds

<i>A. Calibrated parameters</i>		
σ_η 4.8	π 0.06	Ω 29.0
<i>B. Target quantities</i>		
	Data	Simulation
SnM rejections in $\mathcal{S}^\mathcal{E}$	97.9	96.5
SD(ψ) in $\mathcal{S}^\mathcal{E}$	17.2	17.5
$E(\text{sign. } \alpha_s)/E(\text{sign. } \alpha_i)$ in $\mathcal{S}^\mathcal{E}$	0.12	0.12
SnM rejections in $\mathcal{S}^\mathcal{P}$	27.0	30.9
<i>C. Outcome quantities in the set $\mathcal{S}^\mathcal{R}$</i>		
Alpha threshold, $\mathcal{T}_\alpha$		3.8
FM threshold, $\mathcal{T}_\lambda$		3.4
SnM rejections		42.5
False rejections		45.3

We use a global minimization algorithm (particle swarm) to calibrate the statistical framework presented in Section 4. We find the vector of parameters that minimizes the sum of the squared percentage distance of the target quantities. In panel A, we report the calibrated parameters (σ_η and π are expressed as percentages). In panel B, we compare the target quantities obtained from the simulation to the respective quantities from the data. In panel C, we report the adjusted MHT thresholds and the proportion of SnM and false discoveries in the set $\mathcal{S}^\mathcal{R}$.

thresholds for alpha and FM coefficient t -statistics and report their averages across simulations.

We tabulate results in Table 4. Panel A shows the calibrated parameters; panel B compares the target quantities obtained from the simulation and from the data; and panel C reports the MHT adjusted thresholds and the proportion of false discoveries in the set $\mathcal{S}^\mathcal{R}$.

Overall, the simulated economy is very close to the actual data. 96.5% of the signals in the econometrician set $\mathcal{S}^\mathcal{E}$ are deemed as SnM rejections according to the RSW threshold relative to 97.9% in the real data. Similarly, the proportion of SnM rejections in the published research set $\mathcal{S}^\mathcal{P}$ is 30.9%, which is close to 27.0% in the actual data. The residuals from the FM regressions have similar distributions (standard deviation of 17.5% compared to 17.2% in the data). The conversion ratio of signals from stock to portfolios is the same at 0.12.

Moving to the calibrated parameters, we find that the econometrician set $\mathcal{S}^\mathcal{E}$ is endowed with 0.06% of informative signals, confirming the intuitive notion that an overwhelming majority of the randomly constructed strategies are uninformative. For the econometrician (researcher) to estimate a proportion of lucky discoveries of 96.5% (30.9%), we find that the researcher must be 29 times better at drawing informative signals than the econometrician. Thus, our model sustains the hypothesis that researchers do not simply mine the data but are engaged in the process of discovering informative signals.

Ultimately, the calibration allows us to characterize the researcher set $\mathcal{S}^\mathcal{R}$, the proportion of false rejections under SHT, and MHT thresholds based on

the RSW procedure. We find a proportion of false rejections of 45.3%, and thresholds of 3.8 and 3.4 for alpha and FM coefficient t -statistics, respectively.⁷

4.2 Discussion

4.2.1 Thresholds. Our threshold estimates quantitatively confirm the intuition of Harvey, Liu, and Zhu (2016) that accounting for not only the right-tail of the strategies in the published set $\mathcal{S}^P$ but also strategies that do not reach statistical significance in the researcher set $\mathcal{S}^R$ would lead to thresholds higher than three.⁸

The interpretation of our thresholds warrants a few considerations. First, our estimates are for a specific context of empirical asset pricing studies that use a six-factor model and are concerned with abnormally profitable trading strategies. More research is needed to calculate thresholds and develop an understanding of how to use them in the context of corporate finance applications (such as the ones that typically rely on panel regressions). Second, strong priors about the validity of a signal might require special considerations. Harvey (2017) presents a Bayesian approach to deal with such situations. Third, our statistical approach (indeed any MHT procedure) addresses only the multiplicity problem that naturally arises in the research process. These higher thresholds may not be helpful in aligning researchers' personal incentives with the scientific discovery process. Indeed, it is possible that advocating for higher thresholds than those prescribed by SHT might have the unintended consequence of increased data mining. We refer readers to Harvey (2017) for a detailed discussion of this problem and some possible solutions.

Finally, we note that we derive our thresholds by independently applying the RSW procedure to the alpha and FM coefficient t -statistics. However, we use (and suggest that researchers also do so) *both* thresholds to establish which signals are significant. Thus, we effectively create a dual hurdle that trading signals have to surpass. While our statistical framework is cast in a frequentist setting, one can view this dual hurdle in a Bayesian setting (Harvey 2017) as having more skeptical priors than those for a single hurdle. The dual threshold is also very much in the spirit of other procedures that researchers often adopt to reduce the likelihood of data mining, such as testing the same hypotheses with international data.

To consider the statistical effects of such a dual hurdle, we start by looking at the realized FDR (average FDP across simulations). FDP control is stricter

⁷ We also refine the set of stocks used to construct the strategies. It is widely known that anomalies are stronger in small-cap and micro-cap stocks (Fama and French 2008). At portfolio formation, we exclude all stocks with a price less than \$3 and a market capitalization lower than the 20th percentile of the NYSE distribution. This filter also alleviates concerns about transaction costs as well as those about generalizability and relevance (Novy-Marx and Velikov 2016). We find that the proportion of false rejections in this set is 41.7%. Details of these results are available on request.

⁸ Harvey, Liu, and Zhu (2016, p. 38) report that accounting for missing factors increases the 5% BHY threshold from 2.78 to 3.18.

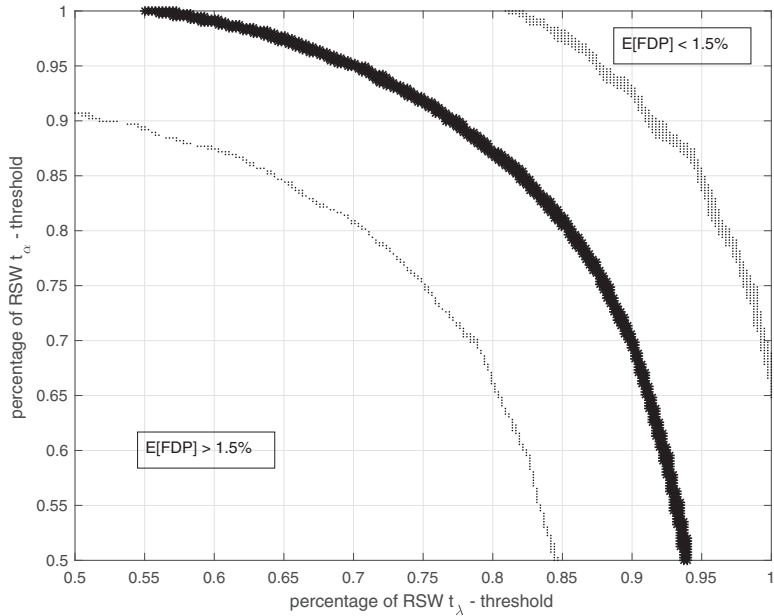


Figure 2
FDR from application of dual RSW thresholds

Using the statistical framework from Section 4 and the baseline parameters from Table 4, we calculate the FDR after applying the dual hurdles on alpha and FM coefficient t -statistics. The solid-black line represents the RSW thresholds for the two t -statistics that attain an average FDR of 1.5% (across simulations). The two light-colored lines show 2-standard-deviations bounds around the average. Below the solid line, the realized FDR is more than 1.5%, and above the line, it is less than 1.5%. The two axes are presented in terms of percentage of the t -statistic thresholds reported in Table 4; the top-right corner corresponds to 3.8 for t_α and 3.4 for t_λ .

than FDR control, so we expect the realized FDR to be lower than the 5% FDP control. We do find that, using the baseline parameters from Table 4, the realized FDR is 0.8% for t_α , 2.2% for t_λ , and 0.2% for the dual hurdles.

While the extremely low FDR of the RSW method achieves the goal of reducing the fraction of false discoveries, it also implies an FDP control of dual hurdles that is lower than what a researcher might desire (e.g., the control used to derive the individual thresholds). Ideally, we would prefer an MHT procedure that ensures an ex ante specified control from the dual hurdles. Absent such a procedure, we offer some heuristics on how to adjust the thresholds to achieve an arbitrary goal. To keep things simple, say that we aim for a realized FDR of 1.5%, where 1.5% is the average realized FDR obtained by individually applying the time-series and cross-sectional thresholds.

Figure 2 shows the combination of thresholds for which the realized FDR is 1.5%. The solid-black line represents for the average (across simulations) and the two light-colored lines show 2-standard-deviations bounds around the average. Below the line, the realized FDR is more than 1.5%, and above the line, it is less than 1.5%. The two axes are presented in terms of percentage of the

thresholds that we report in Table 4, so that the top right corner represents the realized FDR of 0.2% corresponding to the two thresholds of 3.8 and 3.4. Many combinations produce a realized FDR of 1.5%: for example, a dual threshold of 3.1 (80% of 3.8 for t_α) and 2.9 (85% of 3.4 for t_λ). We leave a more explicit analysis of these issues to future research.

4.2.2 False discoveries. The 45.3% estimate is obtained by averaging across the 1,000 simulations, so we can calculate how the simulation error affects the estimate by considering the standard deviation of false rejection percentage across simulations. We find that our estimate has a pretty narrow distribution with a standard deviation of 2.1%. In other words, we can statistically confidently say that our estimate is different from that obtained from the set $\mathcal{S}^P$.

Note that, in general, false rejections in the set $\mathcal{S}^R$ may not be close to SnM rejections. In situations in which the power is not very high, the MHT procedure would not reject all hypotheses that are false under the null and would falsely reject some hypotheses (based on its tolerance level). Low signal-to-noise ratios and limited number of observations (in the time series) make power a significant problem in many finance applications. For example, Andrikogiannopoulou and Papakonstantinou (2019) and Harvey and Liu (2019) show that limited power hinders the application of some MHT methods in mutual fund performance evaluation. Our simulation sidesteps this issue by design as we have a long time series (500 observations).

Our calibration implies that, without any correction for the multiple testing problem, the expected proportion of false rejections would be 45.3% going forward (assuming that the statistical distribution of individual stock returns does not change; researchers do not improve their ability in discovering trading strategies; and MHT does not induce more data mining). In essence, if our model is an accurate representation of researchers' ability to draw informative signals, we should expect to continue to discover abnormally profitable trading strategies with associated t -statistics larger than 1.96. Some of these strategies would be false. If researchers were to adopt MHT thresholds, the percentage of false rejections would be much smaller; possibly as small as the size of the FDP control specified by the MHT procedure.

4.2.3 Relation to out of sample. One alternative approach in the literature to estimate false rejections is out-of-sample studies. If a signal is true under the null (does not produce an alpha), then the signal's predictive ability should be weak in an out-of-sample period. Linnainmaa and Roberts (2018) study the performance of trading strategies in the years before the original study that discovered them. They report that, of the 36 strategies that they study, 20 strategies have insignificant Fama and French (1993) three-factor alphas in the pre-discovery period. This represents a false rejection rate of 55.6%.

McLean and Pontiff (2016) also study out-of-sample performance of 97 trading strategies and find a post publication decay of 58%. They attribute

26% to statistical decay and 32% to publication-informed trading. Under the assumptions that the time-series decay is akin to cross-sectional false rejection rate (i.e., all false strategies decayed to zero and all true strategies did not decay) and that part of the post-publication decay may still reflect false discoveries, their results imply a false rejection rate of between 26% and 58%.

Obvious differences are present in the sample period, the number of strategies, and the performance evaluation method between these studies and ours, but the estimates of false rejections are in the same ballpark. Nevertheless, we return to the issue of different estimates obtained by considering different proportion of lucky rejections in the set $\mathcal{S}^P$ in Section 5.2.

4.2.4 Relation to literature. One of the main difficulties in the application of MHT is the unobservability of the set $\mathcal{S}^R$. Our approach to uncovering the properties of this set is to combine information from the observed sets $\mathcal{S}^E$ and $\mathcal{S}^P$.

Harvey, Liu, and Zhu (2016) also provide an estimate of the set of strategies in the set $\mathcal{S}^R$ by fitting a truncated exponential distribution to the tails of the published strategies in the set $\mathcal{S}^P$. They allow for estimation error in this fitted exponential distribution as well as for the possibility that they may not have captured all the strategies that have been successfully tried. The main difference between our approach and theirs is that instead of matching t -statistics corresponding to a certain frequency (i.e., a certain quantile), we match the frequencies corresponding to certain t -statistics (i.e., the fraction of t -statistics larger than a threshold). Thus, our approach to fitting SnM rejections is akin to fitting the tails of the t -statistic distribution.

The advantage of our approach is that we do not have to necessarily specify the details of the publication process, and we avoid undersampling insignificant strategies. The limitation of our approach is that the shape of the distribution of t -statistics in our simulation may not match the shape of the observed truncated distribution; we match only the fractions of t -statistics larger than 1.96 and larger than the RSW threshold.

Chen (2019) also estimates the truncated part of the distribution of t -statistics (that is present in the set $\mathcal{S}^R$, but not in the set $\mathcal{S}^P$). He fits a variety of models and reports that t -statistic thresholds vary a lot depending on the model. One counter-intuitive aspect is that Chen sometimes finds that researchers never sample strategies from the null of no alpha. Translated into our framework, this means that Ω (researcher's superior ability) is so high that, in the limit, $\Omega \times \pi$ approaches one (Chen's p_0 is close to zero). If a researcher never samples from the null, then, of course, there is no reason for any statistical testing (single or multiple). In such a situation, the t -statistic threshold would be very low, and false rejections would be close to zero, as Chen (2019), and Chen and Zimmermann (2018) report. It is useful to note, however, that even Chen shows that, in situations where his estimated p_0 is further away from zero, the adjusted MHT thresholds are higher than 1.96 and the realized FDR is much higher than 5%.

We believe $p_0 \approx 0$ is not an accurate description of reality. We know from McLean and Pontiff (2016) and Linnainmaa and Roberts (2018) that several strategies do not survive out-of-sample tests. Further, although the Hou, Xue, and Zhang (2018) study is not truly an out-of-sample study, they too find a large fraction of false rejections. Harvey (2017) also highlights biases in the publication process. Finally, some of the research on anomalies is not predicated on theory, at least not on *ex ante* theory.

Our results imply that researchers draw uninformative signals 98% ($1 - \Omega \times \pi \approx 0.98$) of the time. In other words, p_0 is close to 0.98 in our estimates while Chen (2019) suggests that it is much lower. Our high estimate of p_0 is a direct consequence of the very low π . Which of these two estimates is closer to truth might depend on one's priors. Given that our paper is concerned with abnormally profitable strategies, we think that a reasonable prior is based on what we know about the performance of industry practitioners, which we understand to be low (see, e.g., Fama and French 2010; Barras, Scaillet, and Wermers 2010; Busse, Goyal, and Wahal 2010; Chen and Ferson 2019).

5. Robustness Checks

Our estimation of false rejections depends on the assumptions underlying our statistical framework and on the two estimates of SnM rejections (27.0% and 97.9%) that we use. We conduct here a variety of robustness tests.

5.1 Alternative assumptions about stock returns and signals

We first consider the assumption that randomness in the statistical framework described in Section 4 is generated from normal distributions. However, in the data, stock returns exhibit negative skewness and fat tails; also, some factors are prone to extremely negative returns (e.g., momentum crashes; see Daniel and Moskowitz 2016). Moreover, published strategies show some, albeit weak, cross correlation (e.g., Green, Hand, and Zhang 2017 report an average pairwise correlation of 9%).

We first consider Equation (1), which describes three sources of randomness in stock returns: alphas, factor returns, and residuals. We estimate the empirical distribution of the cross-section of alphas in the data. We follow the same procedure of rolling time-series regression, as in Section 4, and compute the time-series averages of the cross-sectional standard deviation, skewness, and kurtosis of estimated stock's alphas. We find skewness of 1.17 and kurtosis of 18.01. Accordingly, we modify the alpha generating process from a normal distribution to a Pearson system with mean zero, standard deviation 1.9%, skewness of 1.17, and kurtosis of 18.01 (see Johnson, Kotz, and Balakrishnan 1994). Results of the calibration are reported in Table 5 under the column entitled "Alpha." While some of the parameters slightly change, we obtain a proportion of false rejections of 45.9%, which is close to the baseline scenario

Table 5
Alternative assumptions about stock returns and signals

	Alpha	Factors	Residuals	$\rho = 9\%$
<i>A. Calibrated parameters</i>				
σ_η	4.2	9.7	7.6	6.2
π	0.06	0.06	0.06	0.08
Ω	28.1	22.1	29.1	22.5
<i>B. Target quantities</i>				
	Data	Simulation		
SnM rejections in $\mathcal{S}^\mathcal{E}$	97.9	96.5	95.8	96.3
SD(ψ) in $\mathcal{S}^\mathcal{E}$	17.2	17.7	17.5	14.7
$E(\text{sign. } \alpha_s)/E(\text{sign. } \alpha_i)$ in $\mathcal{S}^\mathcal{E}$	0.12	0.10	0.13	0.10
SnM rejections in $\mathcal{S}^\mathcal{P}$	27.0	30.0	30.4	28.7
<i>C. Outcome quantities in the set $\mathcal{S}^\mathcal{R}$</i>				
Alpha threshold, $\mathcal{T}_\alpha$	3.8	4.4	3.8	3.8
FM threshold, $\mathcal{T}_\lambda$	3.3	3.5	3.4	3.3
SnM rejections	43.3	44.9	41.2	42.6
False rejections	45.9	46.3	42.8	45.5

We use a global minimization algorithm (particle swarm) to calibrate the statistical framework presented in Section 4. We find the vector of parameters that minimizes the sum of the squared percentage distance of the target quantities. We consider four modifications of the basic assumptions: alternative distribution for alphas, factors and residuals, and signal cross-correlation of 9% (as opposed to zero). In panel A, we report the calibrated parameters. In panel B, we compare the target quantities obtained from the simulation to the respective quantities from the data. In panel C, we report the adjusted MHT thresholds and the proportion of false discoveries.

estimate of 45.3%. The alpha and FM coefficient t -statistic thresholds are also close to the baseline estimates.

Second, instead of simulating factor realizations from a multivariate normal distribution, we bootstrap factor returns by resampling them in stationary blocks with replacement (Politis and Romano 1994) directly from the observed six-factor monthly returns. The bootstrap preserves the higher-order moments and factor crashes observed in the data (and is easier to implement than a simulated system of factors that preserves time-series properties of data counterparts). We find a proportion of false rejections of 46.3%, but higher thresholds at 4.4 and 3.5, for alpha and FM coefficient t -statistics, respectively.

Next, we modify the properties of residuals in Equation (1). We first compute the cross-sectional average of residuals standard deviations, skewness, and kurtosis (again using rolling regressions for each stock) in the real data. We then modify our simulation so that residuals are drawn from a distribution with standard deviation of 14.7%, skewness of 0.83, and kurtosis of 6.38. Recalibrating, we obtain a proportion of false rejections of 42.8%, with critical values of 3.8 and 3.4, for alpha and FM coefficient t -statistics, respectively.

Finally, relying on Green, Hand, and Zhang (2017), we fix the cross-sectional correlation, ρ , between strategies at 9% and repeat the calibration. We find a proportion of false rejections of 45.5% and associated thresholds that are close to the baseline specification at 3.8 and 3.3, for alpha and FM coefficient t -statistics, respectively.

None of these modifications affect our estimate of false rejections in a meaningful way for the following reason. Even though our calibration attempts to match four different quantities in Table 4, the two main drivers are the SnM rejections in the sets $\mathcal{S}^E$ and $\mathcal{S}^P$ (i.e., 97.9% and 27.0%, respectively). Therefore, while some calibrated parameters change, the false rejection proportion is more tightly linked to the two SnM rejections, and relatively independent of the exact specification of randomness in the simulation.

5.2 Alternative definition of SnM rejections in $\mathcal{S}^P$

One essential element of the calibration discussed in section 4.1 is the estimate of SnM rejection in the set of public strategies. We have used the 27% estimate from Harvey, Liu, and Zhu (2016). On the one hand, this estimate has the advantage of being constructed using the pristine set of discoveries presented by the original authors, but, on the other hand, it has the drawback that such discoveries were not based on the same set of procedures.

As a possible alternative, we consider the portfolio returns constructed using the 156 signals that are available from Andrew Chen's Web site (see Chen and Zimmermann 2018). This alternative is not free of its own set of problems. For example, as shown by Green, Hand, and Zhang (2017) and Hou, Xue, and Zhang (2018), there is no guarantee that in a replication sample, and, by changing the evaluation method, the signals would still produce a profitable and statistically significant strategy. Therefore, our replication procedure introduces a bias against discoveries by artificially imposing a threshold for publication. Moreover, even if a signal was profitable in the earlier sample period, it may become unprofitable later because of investor learning (Chordia, Subrahmanyam, and Tong 2014; McLean and Pontiff 2016) and will be deemed unpublishable. The contamination of in-sample (relative to the original study) with out-of-sample data, and the use of a different factor model in the evaluation produces nontrivial consequences for the 156 trading strategies. Only 77 strategies have statistically significant average returns, and of those 57 have statistically significant alphas using the Fama and French (2015) five factors plus the momentum factor. Thus, the set of t -statistics that we can derive from this sample is not as representative of published signals as the one used by Harvey, Liu, and Zhu (2016).

Remaining cognizant of these problems, we compute a SnM rejection rate for those 57 strategies using the same BHY at 5% procedure as that used by Harvey, Liu, and Zhu (2016). We obtain a SnM rejection rate of 50.6%. We repeat the calibration exercise with the new estimate of SnM rejections in $\mathcal{S}^P = 50.6\%$ and report the results in Table 6.

The most immediate difference, relative to the baseline case, is that to allow a higher SnM rejection rate in the set $\mathcal{S}^P$, the researcher must not be as skilled at drawing informative signals. Accordingly, the relative advantage of researcher over the econometrician, Ω , declines from 29.0 in the baseline case to only 13.7 in this experiment. Lower Ω , in turn, implies fewer informative signals in

Table 6
Alternative definition of SnM rejections in $\mathcal{S}^{\mathcal{P}}$

<i>A. Calibrated parameters</i>		
σ_{η} 3.8	π 0.06	Ω 13.7
<i>B. Target quantities</i>		
	Data	Simulation
SnM rejections in $\mathcal{S}^{\mathcal{E}}$	97.9	96.8
SnM rejections in $\mathcal{S}^{\mathcal{P}}$	50.6	48.5
SD(ψ) in $\mathcal{S}^{\mathcal{E}}$	17.2	17.5
$E(\text{sign. } \alpha_i)/E(\text{sign. } \alpha_i)$ in $\mathcal{S}^{\mathcal{E}}$	0.12	0.13
<i>C. Outcome quantities in the set $\mathcal{S}^{\mathcal{R}}$</i>		
Alpha threshold, $\mathcal{T}_{\alpha}$		4.1
FM threshold, $\mathcal{T}_{\lambda}$		3.5
SnM rejections		63.1
False rejections		64.9

We use a global minimization algorithm (particle swarm) to calibrate the statistical framework presented in Section 4. We find the vector of parameters that minimizes the sum of the squared percentage distance of the target quantities. We consider an alternative to the proportion of SnM rejections in $\mathcal{S}^{\mathcal{P}}$ provided by Harvey, Liu, and Zhu (2016): we take the 156 strategies studied by Chen and Zimmerman (2018) and calculate the SnM rejection rate for this set. In panel A, we report the calibrated parameters. In panel B, we compare the target quantities obtained from the simulation to the respective quantities from the data. In panel C, we report the adjusted MHT thresholds and the proportion of false discoveries.

$\mathcal{S}^{\mathcal{R}}$. Combined with a slight increase in the signal-to-noise ratio, σ_{η} , leads to an increase in thresholds (i.e., from 3.8 to 4.1 for t_{α} , and from 3.4 to 3.5 for t_{λ}) and, finally, an increase in the proportion of false rejections to 64.9%. Thus, the impact of higher proportion of SnM rejections in the set of public signals on the estimate of the proportion of false rejections is quantitatively important.

5.3 Alternative specification of SnM rejections in $\mathcal{S}^{\mathcal{E}}$

5.3.1 Different factor models. We consider four alternatives to the Fama and French (2015) five-factor model augmented by the momentum factor: a one-factor model with the market factor (capital asset pricing model, CAPM); the Fama and French (1993) three-factor model (FF3); the Barillas and Shanken (2018) six-factor model (BS); and the Hou, Xue, and Zhang (2015) q -model (HXZ).⁹

As reported in Table 7, we find quite some variation in rejection rates across factor models under SHT. For example, HXZ model rejects the fewest (22.3%), while the BS model rejects the most (35.7%) of the null of zero alphas at conventional statistical levels. Rejection rates for FM coefficient are more

⁹ The control variables in FM regressions correspond to the factor model used. Thus, we do not include any other controls when risk adjusting stock returns by the excess market return. We include size and book-to-market when adjusting stock returns using the FF3 model. In all the other cases, we include size, book-to-market, profitability, asset growth, and 1- and 11-month lagged returns. Panels A1 and A2 of Appendix Table A2 shows summary statistics for alpha and FM t -statistics for these factor models.

Table 7
Different factor models

	Alpha	FM	Alpha and FM
CAPM			
SHT rejections in $\mathcal{S}^{\mathcal{E}}$	23.4	24.8	7.7
RSW rejections in $\mathcal{S}^{\mathcal{E}}$	3.4	8.2	0.6
RSW SnM rejections in $\mathcal{S}^{\mathcal{E}}$	—	—	92.0
Thresholds in $\mathcal{S}^{\mathcal{R}}$	3.8	3.4	—
False rejections in $\mathcal{S}^{\mathcal{R}}$	—	—	40.1
FF3			
SHT rejections in $\mathcal{S}^{\mathcal{E}}$	27.3	24.8	7.8
RSW rejections in $\mathcal{S}^{\mathcal{E}}$	4.9	8.3	0.4
RSW SnM rejections in $\mathcal{S}^{\mathcal{E}}$	—	—	94.3
Thresholds in $\mathcal{S}^{\mathcal{R}}$	3.8	3.4	—
False rejections in $\mathcal{S}^{\mathcal{R}}$	—	—	44.2
BS			
SHT rejections in $\mathcal{S}^{\mathcal{E}}$	35.7	20.6	7.9
RSW rejections in $\mathcal{S}^{\mathcal{E}}$	10.6	5.6	0.7
RSW SnM rejections in $\mathcal{S}^{\mathcal{E}}$	—	—	91.1
Thresholds in $\mathcal{S}^{\mathcal{R}}$	3.8	3.4	—
False rejections in $\mathcal{S}^{\mathcal{R}}$	—	—	46.5
HXZ			
SHT rejections in $\mathcal{S}^{\mathcal{E}}$	22.3	20.6	4.4
RSW rejections in $\mathcal{S}^{\mathcal{E}}$	0.4	4.9	0.1
RSW SnM rejections in $\mathcal{S}^{\mathcal{E}}$	—	—	99.2
Thresholds in $\mathcal{S}^{\mathcal{R}}$	3.8	3.4	—
False rejections in $\mathcal{S}^{\mathcal{R}}$	—	—	45.6

The table reports SHT rejections, RSW rejection rates, and the proportion of SnM rejections for the full set of 2,393,641 signals and all stocks, as well as RSW thresholds and proportion of false rejections obtained from the calibration (see Table A3 in the appendix). The CAPM uses the market factor. FF3 is the Fama and French (1993) three-factor model. BS is the Barillas and Shanken (2015) six-factor model. HXZ is the Hou, Xue, and Zhang (2015) q -model augmented with the momentum factor. In FM regressions, we do not include any other control when risk adjusting stock returns CAPM. We include size and book-to-market when adjusting stock returns using the FF3 model. In all the other cases, we include size, book-to-market, profitability, asset growth, and 1- and 12-month lagged returns. The significance level is 5% for all tests, and we use only the RSW method for MHT control. All rejection rates are expressed as percentages. The sample period is 1972 to 2015.

similar across models. Using FM regressions, the FF3 model has the most rejections (8.3%) and the HXZ model has the least (4.9%). After applying the RSW thresholds, rejections are very low and below 1%.¹⁰

Appendix Table A3 reports detailed results of the calibration of the statistical model for the alternative factor models. Table 7 only reports the RSW thresholds and the proportion of false rejections obtained from the researcher set $\mathcal{S}^{\mathcal{R}}$. Thresholds are all very close to the figures reported in Table 4 across the four possible models.

¹⁰ Taken literally, a fraction close to 0% means that, in our sample of randomly generated strategies, any MHT rejection is likely false when evaluated by the HXZ model. To put it differently, the HXZ model remarkably explains returns of almost all randomly generated returns.

False rejection percentages vary between 40.1% and 46.5%, with CAPM producing the lowest estimate and BS the highest. Although other reasons contribute to the estimate of false rejections, everything else being equal, a higher SnM rejection rate in the set of randomly generated signals, $\mathcal{S}^{\mathcal{E}}$, pushes the magnitude of the estimate of false rejections in the set $\mathcal{S}^{\mathcal{R}}$ higher.

5.3.2 Alternative definition and construction of strategies. First, we repeat our analysis after eliminating signals constructed as $(x_1 - x_2)/x_3$ from the sample set $\mathcal{S}^{\mathcal{E}}$. This smaller set contains 13,748 strategies. Panel B of Appendix Table A2 reports summary statistics for this sample of strategies. Second, instead of decile portfolios, we consider 2×3 portfolios as follows. We sort stocks into two groups based on size with NYSE median determining the cutoff. We independently sort stocks into three groups using as cutoffs the 30th and 70th percentile of the variable of interest. The long-short portfolio is then defined to be long (short) in the average of two size groups for tercile three (one).¹¹ Correspondingly, in the simulation, we sort stocks into three groups using as cutoffs the 30th and 70th percentile of the variable of interest. Third, in FM regressions, we replace raw variables with their ranks that run from zero to one. Such regressions are sometimes used in the literature to remove the effect of outliers (we winsorize the variables at the 0.5 and 99.5 percentiles in the main analysis). Correspondingly, in the simulation, we run FM regressions with ranks.

Table A4 in the appendix reports detailed results from the calibration exercises. Table 8 documents the SHT and MHT rejections, the RSW thresholds, and the proportion of false rejections on these alternate set of strategies. We find that the proportion of false rejections is around 40%, while thresholds remain relatively stable.

6. Conclusion

The finance profession has discovered a number of profitable trading strategies. Given that the CRSP and Compustat data sets have been studied for over four decades, it is likely that the profitability of some of the discovered strategies appears exceptional due to luck. Further, of the many strategies we evaluated in isolation, we only report the profitable ones.

We account for the fact that only a small fraction of the studied strategies is reported and apply MHT to obtain an estimate of the proportion of false discoveries. Even after allowing for the possibility that researchers are much more likely, compared to the econometricians, to oversample strategies that are

¹¹ The 2×3 approach to constructing portfolios is common when constructing factors (see, e.g., Fama and French 1993, 2015). Our paper tries to address the false rejections problem in the context of anomalies, so the question of which 2×3 “factors” are false discoveries is a different question from the one we are trying to answer. Nevertheless, examining our results when following the alternative portfolio sorting approach is interesting.

Table 8
Alternative definitions and construction of strategies

	Alpha	FM	Alpha and FM
Small set of strategies			
SHT rejections in $\mathcal{S}^{\mathcal{E}}$	27.6	19.0	5.2
RSW rejections in $\mathcal{S}^{\mathcal{E}}$	3.7	4.6	0.3
RSW SnM rejections in $\mathcal{S}^{\mathcal{E}}$	86.6	75.6	93.8
Thresholds in $\mathcal{S}^{\mathcal{R}}$	3.8	3.2	—
False rejections in $\mathcal{S}^{\mathcal{R}}$	—	—	40.1
2×3 portfolios			
SHT rejections in $\mathcal{S}^{\mathcal{E}}$	35.5	25.5	11.4
RSW rejections in $\mathcal{S}^{\mathcal{E}}$	16.8	7.4	3.1
RSW SnM rejections in $\mathcal{S}^{\mathcal{E}}$	52.2	71.1	72.7
Thresholds in $\mathcal{S}^{\mathcal{R}}$	3.7	3.3	—
False rejections in $\mathcal{S}^{\mathcal{R}}$	—	—	40.8
FM regressions with ranks			
SHT rejections in $\mathcal{S}^{\mathcal{E}}$	27.5	25.6	10.3
RSW rejections in $\mathcal{S}^{\mathcal{E}}$	8.5	7.5	1.8
RSW SnM rejections in $\mathcal{S}^{\mathcal{E}}$	69.3	70.7	82.1
Thresholds in $\mathcal{S}^{\mathcal{R}}$	3.9	3.4	—
False rejections in $\mathcal{S}^{\mathcal{R}}$	—	—	41.4

The table reports SHT rejections, RSW rejection rates, the proportion of SnM rejections, and RSW thresholds and the proportion of false rejections obtained from the calibration (see Table A4 in the appendix). We consider one alternative to the definition of trading strategies by excluding signals constructed as $(x_1 - x_2)/x_3$. We also consider two alternative ways of constructing trading strategies: sorting stocks into 2×3 groups (instead of deciles) and using ranks of variables (instead of raw variables) in FM regressions. The significance level is 5% for all tests, and we use only the RSW method for MHT control. All rejection rates are expressed as percentages. The sample period is 1972 to 2015.

false under the null, the proportion of false rejections under single hypothesis testing is about 45%.

Appendix

A.1 FWER

We are interested in testing the performance of trading strategies by analyzing the abnormal returns generated by S signals. The test statistic is a t -statistic, or, equivalently, a p -value. The null hypothesis corresponding to each strategy is labeled as H_i . For ease of notation, we will relabel the strategies and order them from the best (highest t -statistic) to the worst (lowest t -statistic). In other words, in the rest of this Appendix, it is assumed that $t_1 \geq t_2 \geq \dots \geq t_S$, or, equivalently, $p_1 \leq p_2 \leq \dots \leq p_S$. The strictest idea in MHT is to try to avoid any false rejections. This translates to controlling the FWER, which is defined as the probability of rejecting even one of the true null hypotheses:

$$\text{FWER} = \text{Prob} (F_1 > 1).$$

Thus, FWER measures the probability of even one false discovery (i.e., rejecting even one true null hypothesis). A testing method is said to control the FWER

at a significance level α if $\text{FWER} \leq \alpha$. Many approaches can be employed to control FWER.

A.1.1 Bonferroni method. All FWER control methods owe their origin to the Bonferroni (1936) adjustment. The logic behind Bonferroni is that if each hypothesis is tested at some significance level α^* , then each hypothesis is erroneously rejected with probability α^* . Because S hypotheses are tested, the expected number of false rejections is $E(F_1) = S \times \alpha^*$. Because $\text{Prob}(F_1 \geq 1) \leq E(F_1)$, to guarantee that $\text{FWER} \leq \alpha$, one should set $\alpha^* = \alpha/S$. Thus, the Bonferroni method rejects H_i if $p_i \leq \alpha/S$. The Bonferroni method is a single-step procedure, because all p -values are compared to a single critical value. As the number S of hypotheses being tested increases, the correction becomes more and more stringent, leading to very high t -statistic thresholds for rejection. Although widely used for its simplicity, the Bonferroni method is very conservative and leads to a loss of power, both of which are considered to be disadvantages of the method. To varying degrees, all procedures that control FWER suffer from the same problem. One of the main reasons for the lack of power is that the Bonferroni method ignores the cross-correlations that are bound to be present in most financial applications.

A.1.2 Holm method. This is a step-down method based on Holm (1979) and works as follows. The procedure starts by checking the most significant hypothesis and works its way down to less significant hypotheses. The null hypothesis H_i is rejected at level α if $p_i \leq \alpha/(S - i + 1)$ for $i = 1, \dots, S$. The intuition for the method is, if one is at stage of the hypothesis i , then hopefully the hypotheses $H_1, H_2, \dots, H_{i-1}$ have been correctly rejected. One can then apply the Bonferroni correction to hypothesis H_i using the number of remaining hypotheses, $S - i + 1$. Compared with that of the Bonferroni method, the Holm method's criterion for the smallest p -value is equally strict at α/S , but it becomes less and less strict for larger p -values. Thus, the Holm method will typically reject more hypotheses and is more powerful than the Bonferroni method. However, because it also does not take into account the dependence structure of the individual p -values, the Holm method is also very conservative.

A.1.3 Bootstrap reality check. Bootstrap reality check (BRC) is based on White (2000). The idea is to estimate the sampling distribution of the largest test statistic taking into account the dependence structure of the individual test statistics, thereby asymptotically controlling FWER. The implementation of the method proceeds as follows. Bootstrap the data using procedure described in Section A.5. For each bootstrapped iteration b , calculate the highest (absolute) t -statistic across all strategies and call it $t_{\max}^{(b)}$, where the superscript b is used to clarify that these t -statistics come from the bootstrap. The critical value is computed as the $(1 - \alpha)$ empirical percentile of B bootstrap iterations values $t_{\max}^{(1)}, t_{\max}^{(2)}, \dots, t_{\max}^{(B)}$. Statistically speaking, BRC can be viewed as a method that

improves on Bonferroni by using the bootstrap to get a less conservative critical value. From an economic point of view, BRC addresses the question of whether the strategy that appears the best in the observed data really beats the benchmark. However, BRC method does not attempt to identify as many outperforming strategies as possible.

A.1.4 StepM method. This method, based on Romano and Wolf (2005), addresses the problem of detecting as many outperforming strategies as possible. The stepwise StepM method is an improvement over the single-step BRC method in very much the same way as the stepwise Holm method improves on the single-step Bonferroni method. The implementation of this procedure proceeds as follows:

1. Consider the set of all S strategies. For each cross-sectional bootstrap iteration, compute the maximum t -statistic, thus obtaining the set $t_{\max}^{(1)}, t_{\max}^{(2)}, \dots, t_{\max}^{(B)}$. Then compute the critical value c_1 as the $(1-\alpha)$ empirical percentile of the set of maximal t -statistics, as in BRC method. Apply now the c_1 threshold to the set of original t -statistics and determine the number of strategies for which the null can be rejected. Say that there are R_1 strategies, for which $t_i \geq c_1$. We have now $S - R_1$ strategies remaining with t -statistics ordered as $t_{R_1+1}, t_{R_1+2}, \dots, t_S$.
2. Consider the set of remaining $S - R_1$ strategies. For each bootstrapped iteration b , calculate the highest (absolute) t -statistic across all remaining strategies. To avoid cluttering the notation, we will use the same symbols as before and call the maximal t -statistics of the b bootstrap iteration across the $S - R_1$ remaining strategies as $t_{\max}^{(b)}$. The critical value c_2 is computed as the $(1-\alpha)$ empirical percentile of B bootstrap iterations values $t_{\max}^{(1)}, t_{\max}^{(2)}, \dots, t_{\max}^{(B)}$. Say that there are R_2 strategies, for which $t_i \geq c_2$, and are, therefore, rejected in this step. After this step, $S - R_1 - R_2$ strategies remain with t -statistics ordered as $t_{R_1+R_2+1}, t_{R_1+R_2+2}, \dots, t_S$.
3. Repeat the procedure until no further strategies are rejected. The StepM critical value for the entire procedure is equal to the critical value of the last step and the number of strategies that are rejected is equal to the sum of the number of strategies that are rejected in each step.

Like the Holm method, the StepM method is a step-down method that starts by examining the most significant strategies. The main advantage of the method is that, because it relies on bootstrap, it is valid under arbitrary dependence structure of the test statistics. As mentioned before, this method will detect many more outperforming strategies than the Bonferroni method or the BRC approach. It is easy to see that the BRC approach amounts to only step one of the above procedure, namely, computing only the critical value c_1 . By continuing the method after the first step, more false null hypotheses can be rejected.

Moreover, typically $c_1 > c_2 > \dots$, so the critical value in StepM method is less conservative than that in BRC approach. Nevertheless, the StepM procedure still asymptotically controls FWER at the significance level of α .

A.2 k -FWER

By relaxing the strict FWER criterion, one can reject more false hypotheses. For instance, k -FWER is defined as the probability of rejecting at least k of the true null hypotheses:

$$k\text{-FWER} = \text{Prob}\{\text{Reject at least } k \text{ of the true null hypothesis}\}.$$

A testing method is said to control for k -FWER at a significance level α if $k\text{-FWER} \leq \alpha$. Testing methods such as Bonferroni and Holm, discussed earlier, can be generalized for k -FWER testing. Please refer to Romano, Shaikh, and Wolf (2008) for further details. Here we discuss only the extension of the StepM method which is known as the k -StepM method.

A.2.1 k -StepM method. The implementation of this procedure proceeds as follows:

1. Consider the set of all S strategies. For each bootstrapped iteration b , calculate the k -highest (absolute) t -statistic across all strategies and call it $t_{k\text{-max}}^{(b)}$, where the superscript b is used to clarify that these t -statistics come from the bootstrap. Compute the critical value c_1 as the $(1-\alpha)$ empirical percentile of B bootstrap iterations values $t_{k\text{-max}}^{(1)}, t_{k\text{-max}}^{(2)}, \dots, t_{k\text{-max}}^{(B)}$. Say that there are R_1 strategies, for which $t_i \geq c_1$, and are, therefore, rejected in this step. After this step, $S - R_1$ strategies remain with t -statistics ordered as $t_{R_1+1}, t_{R_1+2}, \dots, t_S$. Apart from the use of k -max instead of max, this step is identical to the first step of StepM procedure.
2. Consider the set of remaining $S - R_1$ strategies. Call this set *Remain*. Also consider a number $k - 1$ of strategies from the set of already rejected strategies. Call this set *Reject*. Now consider the union of these two sets, $\text{Consider} = \text{Remain} \cup \text{Reject}$. For each bootstrapped iteration b , calculate the k -highest (absolute) t -statistic across all strategies in the set *Consider* and call it $t_{k\text{-max}}^{(b)}$. Compute the $(1-\alpha)$ empirical percentile of B bootstrap iterations values $t_{k\text{-max}}^{(1)}, t_{k\text{-max}}^{(2)}, \dots, t_{k\text{-max}}^{(B)}$. This empirical percentile will depend on which $k - 1$ strategies were included in the set *Reject*. Given that there are $\binom{R_1}{k-1}$ possible ways of choosing $k - 1$ strategies from a set of R_1 strategies, the critical value c_2 is computed as the maximum across all these permutations. Say that there are R_2 strategies, for which $t_i \geq c_2$, and are, therefore, rejected in this step. After this step, $S - R_1 - R_2$ strategies remain with t -statistics ordered as $t_{R_1+R_2+1}, t_{R_1+R_2+2}, \dots, t_S$.

3. Repeat the procedure until no further strategies are rejected. The critical value of the procedure is equal to the critical value of the last step and the number of strategies that are rejected is equal to the sum of the number of strategies that are rejected in each step.

The key innovation in the k -StepM procedure is in the inclusion of (some of the) rejected strategies while calculating subsequent critical values (c_2 and thereafter). The intuition is as follows. Remember that ideally we want to calculate the empirical critical value from the set of strategies that are true under the null hypothesis. This set is unknown in practice. However, we can use the results of the first step to arrive at this set. The set *Remain* of remaining strategies that have not (yet) been rejected is an obvious candidate for strategies that are true under the null. If we are in the second step of the procedure, it stands to reason that the first step was not able to control k -FWER. In other words, less than k true null hypotheses were rejected in the first step. Let's say that number is in fact $k - 1$. Obviously, we do not know with precision which $k - 1$ true nulls have been rejected among the many strategies rejected in the first step. Therefore, to be cautious, Romano, Shaikh, and Wolf (2008) suggest looking at all possible combinations of $k - 1$ rejected hypotheses from the set *Reject*.

A.3 False Discovery Rate

A multiple testing method is said to control FDR at level δ if $\text{FDR} \equiv E(\text{FDP}) \leq \delta$. FDR is already an expectation, so controlling for FDR does not need additional specification of probabilistic significance level. One of the earliest methods to controlling FDR is by Benjamini and Hochberg (1995) and proceeds in a stepwise fashion as follows. Assuming as before that the individual p -values are ordered from the smallest to largest, and defining:

$$j^* = \max \left\{ j : p_j \leq \left(\frac{j}{S} \right) \delta \right\},$$

one rejects all hypotheses $H_1, H_2, \dots, H_{j^*}$ (i.e., j^* is the index of the largest p -value among all hypotheses that are rejected). This is a step-up method that starts with examining the least significant hypothesis and moves up to more significant test statistics. Benjamini and Hochberg (1995) show that their method controls FDR if the p -values are mutually independent. Benjamini and Yekutieli (2001) show that a more general control of FDR under a more general dependence structure of p -values can be achieved by replacing the definition of j^* with

$$j^* = \max \left\{ j : p_j \leq \left(\frac{j}{S \times C_S} \right) \delta \right\},$$

where the constant $C_S = \sum_{i=1}^S 1/i \approx \log(S)$. However, the Benjamini and Yekutieli (2001) method is less powerful than that of Benjamini and Hochberg (1995).

A.4 False Discovery Proportion

A multiple testing method is said to control FDP at proportion γ and level α if $\text{Prob}(\text{FDP} > \gamma) \leq \alpha$. Here we discuss the extension of the StepM procedure developed by Romano and Wolf (2007).

A.4.1 FDP-StepM method. The StepM procedure for control of FDP is as follows:

1. Let $k = 1$.
2. Apply the k -StepM method and denote by R_k the number of hypotheses rejected.
3. If $R_k < k/\gamma - 1$, then stop. Else, let $k = k + 1$ and return to step 2.

The FDP-StepM method is, thus, a sequence of k -StepM procedures. The intuition of applying an increasing series of k 's is as follows. Consider controlling FDP at proportion $\gamma = 10\%$. We start by applying the 1-StepM method. Denote by R_1 the number of strategies rejected at this stage. The basic 1-StepM procedure controls for FWER, so we can be confident that no false rejections have occurred so far, which in turn also implies that FDP also has been controlled. Consider now the issue of rejecting the strategy H_{R_1+1} , the next most significant strategy (recall that StepM is a step-down procedure). Rejection of H_{R_1+1} , if the null of this strategy is true, renders the false discovery proportion to be equal to $1/(R_1 + 1)$. We are willing to tolerate 10% of false rejections, so we would be willing to tolerate rejecting this strategy if $1/(R_1 + 1) < 0.1$, which is true if $R_1 > 9$. Thus, if $R_1 < 9$ the procedure would stop at the first step. Alternatively, if $R_1 > 9$, the procedure would continue with the 2-StepM method, which by design should not reject more than one true hypothesis. Besides the fact that the FDP-StepM method allows the researcher to directly control FDP, one other big advantage of this method for us is that it accounts for an arbitrary dependence structure in the data and, therefore, in the individual p -values.

A.5 Bootstrap Method

Some of the methods described above rely on a bootstrap. We describe the details of the bootstrap in this section. This approach, inspired by Kosowski et al. (2006) and Fama and French (2010) and used by Yan and Zheng (2017), relies on bootstrapping the cross-section of fund returns through time thereby preserving the cross-sectional dependence structure in strategy returns and ultimately their alpha estimates. To bootstrap under the null, say of zero alpha, we first subtract the factor-model alpha from the monthly portfolio returns. Each bootstrap run is a random sample (with replacement) of the alpha-adjusted

returns and the factors over the sample period.¹² To preserve the cross-sectional correlation, we apply the same bootstrap draw to all portfolios and to the factors. To preserve possible autocorrelation in the return structure, we construct the stationary bootstrap of Politis and Romano (1994) by drawing random blocks with an average length of 6 months. Because of the computational constraints imposed by the large scale of our exercise we limit the exercise to 1,000 bootstrap samples. For each bootstrap run we obtain the portfolio alphas and their t -statistics under the null of zero alpha. This bootstrapped distribution of t -statistics is used in the MHT methods that need it. We conduct a similar experiment for FM coefficients. In particular, for each signal variable we start by subtracting the average from the time series of the FM coefficient thus obtaining a time series of adjusted coefficients under the null of no explanatory power. We then bootstrap 1,000 times the time series of pseudo-coefficients and calculate the means and t -statistics for each bootstrap iteration. This generates the bootstrapped distribution of t -statistics of FM coefficients under null.

¹² Romano and Wolf (2005) suggest bootstrapping the data without imposing the null but computing critical values by “recentering” the distribution of bootstrapped statistics based on data statistics (see their algorithms 3.2 and 4.2). This alternative approach yields equivalent result to bootstrapping under the null in our context.

Table A1
Basic variables used to construct trading strategies

#	Short	Long	#	Short	Long
1	aco	Current assets, other total	61	ebit	Earnings before interest and taxes
2	acox	Current assets, other sundry	62	ebitda	Earnings before interest
3	act	Current assets, total	63	emp	Employees
4	am	Amortization of intangibles	64	epsfi	Earnings per share (diluted) including extraordinary items
5	ao	Assets other	65	epsfx	Earnings per share (diluted) excluding extraordinary items
6	aox	Assets other sundry	66	epspi	Earnings per share (basic) including extraordinary items
7	ap	Accounts payable trade	67	epspx	Earnings per share (basic) excluding extraordinary items
8	aqc	Acquisitions	68	esub	Equity in earnings unconsolidated subsidiaries
9	aqi	Acquisitions income contribution	69	esubc	Equity in net loss earnings
10	aqs	Acquisitions sales contribution	70	fca	Foreign exchange income (loss)
11	at	Assets total	71	fopo	Funds from operations other
12	bkvtps	Book value per share	72	gp	Gross profit (loss)
13	caps	Capital surplus-share premium reserve	73	ib	Income before extraordinary items
14	capx	Capital expenditures	74	ibadj	Income before extraordinary items adjusted for common stock equivalents
15	capxv	Capital expend property, plant and equipment schd V	75	ibc	Income before extraordinary items (cash flow)
16	ceq	Common-ordinary equity total	76	ibcom	Income before extraordinary items available for common
17	ceql	Common equity liquidation value	77	icapt	Invested capital total
18	ceqt	Common equity tangible	78	idit	Interest and related income total
19	ch	Cash	79	intan	Intangible assets total
20	che	Cash and short-term investments	80	intc	Interest capitalized
21	chech	Cash and cash equivalents increase-(decrease)	81	invfg	Inventories finished goods
22	cogs	Cost of goods sold	82	invrm	Inventories raw materials
23	cshfd	Common shares used to calc earnings per share fully diluted	83	invrt	Inventories total
24	csho	Common shares outstanding	84	invwip	Inventories work in process
25	cshpri	Common shares used to calculate earnings per share basic	85	itcb	Investment tax credit (balance sheet)
26	cshr	Common-ordinary shareholders	86	itci	Investment tax credit (income account)
27	csk	Common-ordinary stock (capital)	87	ivaec	Investment and advances equity
28	cskev	Common stock-carrying value	88	ivao	Investment and advances other
29	cske	Common stock equivalents dollar savings	89	ivch	Increase in investments

(Continued)

Table A1
(Continued)

#	Short	Long	#	Short	Long
30	dc	Deferred charges	90	ivst	Short-term investments total
31	dclo	Debt capitalized lease obligations	91	lco	Current liabilities other total
32	depsik	Convertible debt and preferred stock	92	lcox	Current liabilities other sundry
33	devsr	Debt senior convertible	93	let	Current liabilities total
34	dcvsub	Debt subordinated convertible	94	lifr	LIFO reserve
35	dcvt	Debt convertible	95	lifrp	LIFO reserve prior
36	dd	Debt debentures	96	lo	Liabilities other total
37	dd1	Long-term debt due in one year	97	lse	Liabilities and stockholders equity total
38	dd2	Debt due in 2nd year	98	lt	Liabilities total
39	dd3	Debt due in 3rd year	99	mib	Noncontrolling interest (balance sheet)
40	dd4	Debt due in 4th year	100	mibt	Noncontrolling interest total balance sheet
41	dd5	Debt due in 5th year	101	mi	Noncontrolling interest (income account)
42	dle	Debt in current liabilities total	102	mrc1	Rental commitments minimum 1st year
43	dlts	Long-term debt issuance	103	mrc2	Rental commitments minimum 2nd year
44	dlto	Other Long-term debt	104	mrc3	Rental commitments minimum 3rd year
45	dltp	Long-term debt tied to prime	105	mrc4	Rental commitments minimum 4th year
46	dltr	Long-term debt reduction	106	mrc5	Rental commitments minimum 5th year
47	dltt	Long-term debt total	107	mrcr	Rental commitments minimum 5 year total
48	dm	Debt mortgages other secured	108	msa	Marketable securities adjustment
49	dn	Debt notes	109	ni	Net income (loss)
50	do	Discontinued operations	110	niadj	Net income adjusted for common-ordinary stock (capital) equivalents
51	dp	Depreciation and amortization	111	nopi	Nonoperating income (expense)
52	dpact	Depreciation, depletion and amortization (accumulated)	112	nopio	Nonoperating income (expense) other
53	dpc	Depreciation and amortization (cash flow)	113	np	Notes payable short-term borrowings
54	dpvieb	Depreciation (accumulated) ending balance (schedule VI)	114	ob	Order backlog
55	ds	Debt-subordinated	115	oiadp	Operating income after depreciation
56	dv	Cash dividends (cash flow)	116	oiadp	Operating income before depreciation
57	dvc	Dividends common-ordinary	117	pi	Pretax income
58	dvp	Dividends preferred-preference	118	ppgst	Property, plant and equipment total (gross)
59	dvpa	Preferred dividends in arrears	119	ppnet	Property, plant and equipment total (net)
60	dv1	Dividends total	120	ppveeb	Property, plant, and equipment ending balance (Schedule V)

(Continued)

Table A1
(Continued)

#	Short	Long	#	Short	Long
121	pskcc	Purchase of common and preferred stock	154	txfo	Income taxes foreign
122	pskc	Preferred-preference stock (capital) total	155	txp	Income taxes payable
123	pskc	Preferred stock convertible	156	txr	Income tax refund
124	pskl	Preferred stock liquidating value	157	txs	Income taxes state
125	pskn	Preferred-preference stock nonredeemable	158	txt	Income taxes total
126	pskr	Preferred-preference stock redeemable	159	txw	Excise taxes
127	pskrv	Preferred stock redemption value	160	wcap	Working capital (balance sheet)
128	re	Retained earnings	161	xacc	Accrued expenses
129	rea	Retained earnings restatement	162	xad	Advertising expense
130	reajo	Retained earnings other adjustments	163	xi	Extraordinary items
131	recco	Receivables current other	164	xido	Extraordinary items and discontinued operations
132	recd	Receivables estimated doubtful	165	xidoc	Extraordinary items and discontinued operations (cash flow)
133	rect	Receivables total	166	xint	Interest and related expense total
134	recta	Retained earnings cumulative translation adjustment	167	xlr	Staff expense total
135	rectr	Receivables trade	168	xopr	Operating expenses total
136	reuna	Retained earnings unadjusted	169	xpp	Prepaid expenses
137	revt	Revenue total	170	xpr	Pension and retirement expense
138	sale	Sales-turnover (net)	171	xrd	Research and development expense
139	seq	Stockholders equity parent	172	xrdp	Research development prior
140	siv	Sale of investments	173	xrent	Rental expense
141	spl	Special items	174	xsga	Selling, general and administrative expense
142	sppc	Sale of property	175	ret1	1m past return
143	ssk	Sale of common and preferred stock	176	ret3	3m past return
144	tlcf	Tax loss carry forward	177	ret6	6m past return
145	tsk	Treasury stock total (all capital)	178	ret9	9m past return
146	tskc	Treasury stock common	179	ret12	1y past return
147	tskn	Treasury stock number of common shares	180	size	Market capitalization
148	txc	Income taxes current	181	price	Price
149	txdb	Deferred taxes (balance sheet)	182	volume	Traded volume
150	txdc	Deferred taxes (cash flow)	183	dvolume	Dollar traded volume
151	txdi	Income taxes deferred	184	turn	Turnover
152	txdic	Deferred taxes and investment tax credit	185	vol	1y return volatility

Table A2
Descriptive statistics of portfolio raw returns on trading strategies: Subsamples and different factor models

	Mean	Median	SD	Min	Max	% <i>t</i> > 1.96	% <i>t</i> > 2.57
<i>A1. Alpha <i>t</i>-statistics for different factor models</i>							
CAPM	−0.09	−0.12	1.64	−7.53	7.88	23.4	12.2
FF3	−0.24	−0.26	1.76	−8.15	8.69	27.4	14.8
BS	−0.29	−0.29	2.09	−8.23	7.96	35.7	23.1
HXZ	−0.19	−0.18	1.58	−7.30	7.78	22.3	11.1
<i>A2. FM <i>t</i>-statistics for different factor models</i>							
CAPM	0.14	0.03	1.97	−11.55	11.55	24.8	14.4
FF3	−0.02	−0.07	1.84	−9.27	8.58	24.8	14.7
BS	0.07	0.03	1.66	−8.58	8.33	20.6	11.6
HXZ	0.05	0.01	1.64	−8.47	7.92	20.6	11.4
<i>B. Small set of strategies</i>							
Return	−0.08	−0.10	1.21	−7.04	6.72	10.6	3.7
Alpha	−0.19	−0.19	1.62	−7.80	9.01	23.2	11.9
Alpha 2×3	0.14	0.10	2.25	−7.38	7.97	35.1	24.1
FM	0.07	0.03	1.64	−8.63	8.12	20.0	11.2
FM rank	−0.38	−0.31	1.73	−5.46	5.95	28.0	16.1

This table reports the cross-sectional mean, median, standard deviation, minimum, and maximum of the *t*-statistics of monthly average return, alpha, and FM coefficients as in Table 3 but for subsamples and different factor models. Panel A uses all stocks and all strategies but uses the different factor models described in Table 7. Panel B is for the subsample of the main set of strategies that does not contain portfolio returns constructed using *Ratio of three* signals. The subsample comprises 13,748 strategies and uses the Fama and French (2015) five-factor model augmented with the momentum factor. The row entitled “Alpha 2×3” sorts stocks into 2×3 groups (instead of deciles), and the row entitled “FM rank” uses ranks of variables (instead of raw variables) in FM regressions. The sample period is 1972 to 2015.

Table A3
Alternative factor models: Calibration

	CAPM		FF3		BS		HXZ		
<i>A. Calibrated parameters</i>									
σ_η	3.0		3.1		7.4		3.2		
π	0.13		0.09		0.10		0.03		
Ω	16.1		21.1		17.1		56.9		
<i>B. Target quantities</i>									
		Data	Sim	Data	Sim	Data	Sim	Data	Sim
SnM rejections in $\mathcal{S}^\mathcal{E}$		92.0	91.9	94.3	94.2	91.1	94.1	99.2	98.6
SD(ψ) in $\mathcal{S}^\mathcal{E}$		17.1	17.5	17.1	17.5	17.2	17.6	17.2	17.7
$E(\text{sign. } \alpha_s)/E(\text{sign. } \alpha_i)$ in $\mathcal{S}^\mathcal{E}$		0.20	0.20	0.14	0.16	0.12	0.12	0.15	0.11
SnM rejections in $\mathcal{S}^\mathcal{P}$		27.0	26.7	27.0	29.9	27.0	32.0	27.0	32.5
<i>C. Outcome quantities in the set $\mathcal{S}^\mathcal{R}$</i>									
Alpha threshold, $\mathcal{T}_\alpha$		3.8		3.8		3.8		3.8	
FM threshold, $\mathcal{T}_\lambda$		3.4		3.4		3.4		3.4	
SnM rejections		38.0		42.0		44.1		42.2	
False rejections		40.1		44.2		46.5		45.6	

We use a global minimization algorithm (i.e., particle swarm) to calibrate the statistical framework presented in Section 2 and 4. We find the vector of parameters that minimizes the sum of the squared percentage distance of the target quantities. We consider four alternative factor models: CAPM, FF3, BS, and HXZ. In panel A, we report the calibrated parameters. In panel B, we compare the target quantities obtained from the simulation to the respective quantities from the data. In panel C, we report the adjusted MHT thresholds and the proportion of false discoveries.

Table A4
Alternative definitions and construction of strategies: Calibration

	Small set of strategies	2×3 portfolios		FM regressions with ranks		
<i>A. Calibrated parameters</i>						
σ_η	5.6		19.3		13.9	
π	0.10		0.75		0.31	
Ω	21.1		3.4		5.1	
<i>B. Target quantities</i>						
		Data	Sim	Data	Sim	Data
SnM rejections in $\mathcal{S}^\mathcal{E}$		93.8	93.7	72.7	71.0	82.2
SD(ψ) in $\mathcal{S}^\mathcal{E}$		14.2	15.6	14.2	15.6	14.3
$E(\text{sign. } \alpha_s)/E(\text{sign. } \alpha_i)$ in $\mathcal{S}^\mathcal{E}$		0.11	0.12	0.08	0.08	0.13
SnM rejections in $\mathcal{S}^\mathcal{P}$		27.0	28.3	27.0	28.2	27.0
<i>C. Outcome quantities in the set $\mathcal{S}^\mathcal{R}$</i>						
Alpha threshold, $\mathcal{T}_\alpha$		3.8			3.7	
FM threshold, $\mathcal{T}_\lambda$		3.2			3.3	
SnM rejections		38.2			36.7	
False rejections		40.9			40.8	

We use a global minimization algorithm (particle swarm) to calibrate the statistical framework presented in Section 4. We find the vector of parameters that minimizes the sum of the squared percentage distance of the target quantities. We consider one alternative to the definition of trading strategies by excluding signals constructed as $(x_1 - x_2)/x_3$. We also consider two alternatives ways of constructing trading strategies: sorting stocks into 2×3 groups (instead of deciles) and using ranks of variables (instead of raw variables) in FM regressions. In panel A, we report the calibrated parameters. In panel B, we compare the target quantities obtained from the simulation to the respective quantities from the data. In panel C, we report the adjusted MHT thresholds and the proportion of false discoveries.

References

- Andrikogiannopoulou, A., and F. Papakonstantinou. (2019). Reassessing false discoveries in mutual fund performance: Skill, luck, or lack of power? *Journal of Finance* 74:2667–88.
- Barillas, F., and J. Shanken. (2018). Comparing asset pricing models. *Journal of Finance* 73:715–54.
- Barras, L., O. Scaillet, and R. Wermers. 2010. False discoveries in mutual fund performance: Measuring luck in estimated alphas. *Journal of Finance* 65:179–216.
- Benjamini, Y., and Y. Hochberg. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society, Series B (Methodological)* 57:289–300.
- Benjamini, Y., and D. Yekutieli. (2001). The control of the false discovery rate in multiple testing under dependency. *Annals of Statistics* 29:1165–88.
- Bonferroni, C. 1936. *Teoria statistica delle classi e calcolo delle probabilità*. Florence, Italy: Libreria Internazionale Seeber.
- Brennan, M., T. Chordia, and A. Subrahmanyam. 1998. Alternative factor specifications, security characteristics, and the cross-section of expected stock returns. *Journal of Financial Economics* 49:345–73.
- Busse, J. A., A. Goyal, and S. Wahal. 2010. Performance and persistence in institutional investment management. *Journal of Finance* 65:765–90.
- Carhart, M. 1997. On persistence in mutual fund performance. *Journal of Finance* 52:57–82.
- Chang, A., and P. Li. (2018). Is economics research replicable? Sixty published papers from thirteen journals say “Often Not.” *Critical Finance Review*. Advance Access published October 1, 2018, 10.1561/104.00000053.
- Chen, A. 2019. Do t-stat hurdles need to be raised? Identification of publication bias in the cross-section of stock returns. Working Paper, Federal Reserve Board.
- Chen, A., and T. Zimmermann. (2019). Publication bias and the cross-section of stock returns. *Review of Asset Pricing Studies*. Advance Access published November 27, 2019, 10.1093/rapstu/raz011.
- Chen, Y., and W. Ferson. (2019). How many good and bad funds are there, really? In *Handbook of financial economics, mathematics, statistics, and technology*, eds. C. F. Lee and J. C. Lee. Singapore: World Scientific Press.
- Chordia, T., A. Subrahmanyam, and Q. Tong. 2014. Have capital market anomalies attenuated in the recent era of high liquidity and trading activity? *Journal of Accounting and Economics* 58:41–58.
- Conrad, J., M. Cooper, and G. Kaul. 2003. Value versus glamour. *Journal of Finance* 58:1969–95.
- Daniel, K., and T. Moskowitz. (2016). Momentum crashes. *Journal of Financial Economics* 122:221–47.
- Dewald, W., J. Thursby, and R. Anderson. 1986. Replication in empirical economics: The Journal of Money, Credit, and Banking project. *American Economic Review* 76:587–630.
- Fama, E., and K. French. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics* 33:3–56.
- . 2008. Dissecting anomalies. *Journal of Finance* 63:1653–78.
- . 2010. Luck versus skill in the cross-section of mutual fund returns. *Journal of Finance* 65:1915–47.
- . 2015. A five-factor asset pricing model. *Journal of Financial Economics* 116:1–22.
- Fama, E., and J. MacBeth. (1973). Risk, return and equilibrium: Empirical tests. *Journal of Political Economy* 81:607–36.
- Foster, D., T. Smith, and R. Whaley. 1997. Assessing goodness-of-fit of asset pricing models: The distribution of the maximal R². *Journal of Finance* 52:591–607.

- Green, J., J. Hand, and F. Zhang. 2017. The characteristics that provide independent information about average U.S. monthly stock returns. *Review of Financial Studies* 30:4389–436.
- Genovese, C., and L. Wasserman. (2006). Exceedance control of the false discovery proportion. *Journal of the American Statistical Association* 101:1408–17.
- Harvey, C. 2017. The scientific outlook in financial economics. *Journal of Finance* 72:1399–440.
- Harvey, C., and Y. Liu. (2019). False (and missed) discoveries in financial economics. *Journal of Finance*.
- Harvey, C., Y. Liu, and A. Saretto. 2019. An evaluation of alternative multiple testing methods for finance application. Working Paper, Duke University.
- Harvey, C., Y. Liu, and H. Zhu. 2016. ... and the cross-section of expected returns. *Review of Financial Studies* 29:5–68.
- Holm, S. 1979. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics* 6:65–70.
- Hou, K., C. Xue, and L. Zhang. 2015. Digesting anomalies: An investment approach. *Review of Financial Studies* 28:650–705.
- . 2020. Replicating anomalies. *Review of Financial Studies* 33:2019–2133.
- Ioannidis, J. 2005. Why most published research findings are false. *PLoS Medicine* 2:696–701.
- Johnson, N., S. Kotz, and N. Balakrishnan. 1994. *Continuous univariate distributions I*, 2nd ed. New York: John Wiley & Sons.
- Karolyi, A., and B.-C. Kho. 2004. Momentum strategies: Some bootstrap tests. *Journal of Empirical Finance* 11:509–36.
- Kennedy, J., and R. Eberhart. (1995). Particle swarm optimization. *Proceedings of the IEEE International Conference on Neural Networks*, 1942–48. Piscataway, NJ: IEEE Publishing.
- Kosowski, R., A. Timmermann, R. Wermers, and H. White. 2006. Can mutual fund “stars” really pick stocks? New evidence from a bootstrap analysis. *Journal of Finance* 61:2551–95.
- Leamer, E. 1983. Let’s take the con out of econometrics. *American Economic Review* 73:31–43.
- Linnainmaa, J., and M. Roberts. (2018). The history of the cross-section of stock returns. *Review of Financial Studies* 31:2606–49.
- Lo, A., and C. MacKinlay. (1990). Data-snooping biases in tests of financial asset pricing models. *Review of Financial Studies* 3:431–67.
- Martin, I., and S. Nagel. (2019). Market efficiency in the age of big data. Working Paper, London School of Economics.
- McCullough, B., and H. Vinod. (2003). Verifying the solution from a nonlinear solver: A case study. *American Economic Review* 93:873–92.
- McLean, D., and J. Pontiff. (2016). Does academic research destroy stock return predictability? *Journal of Finance* 71:5–32.
- Novy-Marx, R., and M. Velikov. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies* 29:104–47.
- Politis, D., and J. Romano. (1994). The stationary bootstrap. *Journal of the American Statistical Association* 89:1303–13.
- Romano, J., A. Shaikh, and M. Wolf. 2008. Formalized data snooping based on generalized error rates. *Econometric Theory* 24:404–47.
- Romano, J., and M. Wolf. (2005). Stepwise multiple testing as formalized data snooping. *Econometrica* 73:1237–82.

- . 2007. Control of generalized error rates in multiple testing. *Annals of Statistics* 35:1378–408.
- Schwert, W. 2003. Anomalies and market efficiency. In *Handbook of the economics of finance*, vol. 1B, eds. G. M. Constantinides, M. Harris, and R. M. Stulz, 939–74. Amsterdam, the Netherlands: Elsevier.
- Sullivan, R., A. Timmermann, and H. White. 1999. Data-snooping, technical trading rule performance, and the bootstrap. *Journal of Finance* 54:1647–91.
- Yan, X., and L. Zheng. (2017). Fundamental analysis and the cross-section of stock returns: A data-mining approach. *Review of Financial Studies* 30:1382–423.
- White, H. 2000. A reality check for data snooping. *Econometrica* 68:1097–126.